Posterior abdominal wall and retroperitoneum

CHAPTER 62

DEFINITIONS, BOUNDARIES AND CONTENTS

The posterior abdominal wall does not have an agreed uniform definition. It represents the posterior boundary of the abdominal cavity. In common with the anterior and lateral abdominal walls, it is composed of several layers (skin, superficial fascia, muscle, extraperitoneal fat/fascia and parietal peritoneum). The vertebral column and paravertebral muscles of the back are part of the posterior abdominal wall, but are usually not considered in this context. The posterior abdominal wall is continuous with the posterior thoracic wall at the posterior attachment of the diaphragm, with the posterior wall of the pelvis inferiorly, and with the anterolateral abdominal wall laterally.

The retroperitoneum is a three-dimensional compartment bounded anteriorly by the parietal peritoneum of the posterior abdominal wall and superiorly by the diaphragm. Inferiorly and laterally, it is continuous with the extraperitoneal connective tissues lining the pelvis and anterolateral abdominal wall, respectively. The posterior boundary of the retroperitoneum is variably defined, depending on whether the psoas muscles and quadratus lumborum are regarded as contents or boundaries of this compartment.

The posterior abdominal wall and the retroperitoneum are related but different concepts. The retroperitoneum is a compartment and a three-dimensional space, a concept invoked when localization, spread and containment of disease are primary considerations. The posterior abdominal wall is a part of the parietal coverings of abdominal viscera, a concept invoked when considering posteriorly directed surgery from within the abdomen, posterolateral access to retroperitoneal organs (e.g. nephrectomy or percutaneous endoscopic renal access), or else in contradistinction to the anterior abdominal wall. Advances in cross-sectional imaging have promoted a better understanding of the anatomy of the retroperitoneum, particularly its compartmental anatomy, and with it the retroperitoneum has become the more prevalent clinical concept outside of specific surgical circumstances.

The retroperitoneum houses the paired viscera originating from the embryonic extracoelomic space: the kidneys and ureters (Ch. 74), and the suprarenal glands (Ch. 71) and their vessels and nerves. It also houses the unpaired derivatives of the embryonic intracoelomic gut tube secondarily retroperitonealized: the duodenum (Ch. 65), the pancreas (Ch. 69), and the ascending and descending colon (Ch. 66) and their vessels and nerves.

Attached to the retroperitoneum are the mesenteries of the small bowel (Ch. 65), transverse and sigmoid colons (Ch. 66), and the peritoneal ligaments of the liver (Ch. 67) and spleen (Ch. 70).

The retroperitoneum contains the abdominal aorta and branches; the inferior vena cava and tributaries; the origins of the azygos and hemiazygos veins; the pre- and para-aortic (lumbar) lymph nodes, cisterna chyli and the origin of the thoracic duct (Ch. 56); the diaphragmatic crura (Ch. 55); the lumbar plexus and lumbosacral trunk; and the autonomic plexuses of the abdomen (Ch. 59).

SKIN AND SOFT TISSUES

The skin of the posterior abdominal wall is supplied by musculocutaneous branches of the lumbar arteries and veins, and innervated by the dorsal rami of the lower thoracic and lumbar spinal nerves. However, the cutaneous distribution of the dorsal rami of L4 and L5 is variable and controversial (Lee et al 2008).

In the muscular posterior abdominal wall, intermuscular fascial layers delineate compartments and provide the conceptual anatomical framework. In the retroperitoneum, the compartments (the clinical synonym is 'spaces') and their interrelationships provide the conceptual framework, particularly in cross-sectional anatomy, with the dividing fasciae often hard to visualize unless outlined by disease processes.

THORACOLUMBAR FASCIA

The thoracolumbar fascia is composed of a complex arrangement of multiple fascial layers that is most prominent at the caudal end of the lumbar spine (Ch. 43) (Willard et al 2012). In the lumbar region, it is often described as having three layers (**Figs 62.1–62.3**, see Fig. 43.6). The posterior layer is attached medially to the spines of the lumbar vertebrae and to the supraspinous ligament; it has a superficial lamina (the aponeurosis of latissimus dorsi) and a deep lamina that covers the posterior surface of the paraspinal muscles. The middle layer is attached medially to the tips of the transverse processes of the lumbar vertebrae and extends laterally behind quadratus lumborum; inferiorly, it attaches to the iliac crest, and superiorly to the lower border of the twelfth rib. The anterior layer covers the anterior surface of quadratus lumborum and is attached medially to the transverse processes of the lumbar vertebrae behind psoas major. Laterally, it fuses with the transversalis fascia and the aponeurosis of transversus abdominis. Inferiorly, it is attached to the iliolumbar ligament and adjoining iliac crest. Superiorly, it is attached to the inferior border of the twelfth rib and extends to the transverse process of the first lumbar vertebra, forming the lateral arcuate ligament of the diaphragm. The posterior and middle layers of the thoracolumbar fascia fuse at the lateral margin of the paraspinal muscles (the so-called 'lateral raphe'), thereby enclosing the paraspinal muscles in an osteofascial compartment. Although contained in layers of thoracolumbar fascia, the paraspinal muscles are conceptualized as part of 'the back'. The aponeurosis of transversus abdominis fuses with both the anterior layer of thoracolumbar fascia at the lateral margin of quadratus lumborum and with the lateral raphe behind quadratus lumborum.

ILIOPSOAS FASCIA

Psoas fascia

A relatively dense layer of fascia covers the anterior surface of psoas major. Medially, it is continuous with the attachments of the muscle to the transverse processes and bodies of the lumbar vertebrae and the tendinous arches. Laterally, the fascia blends with the fascia over quadratus lumborum above and is continuous with the iliac fascia below. Superiorly, the fascia merges with the medial arcuate ligament, while, inferiorly, it extends down into the thigh around the iliopsoas tendon. The psoas fascia separates the anterior surface of the muscle and the lumbar plexus within it from the retroperitoneal structures lying anteriorly. A psoas abscess, which can arise by haematogenous seeding or from direct extension of infection from adjacent structures (e.g. the spine, kidneys or colon), often tracks along the muscle within the fascia, and, rarely, may 'point' in the groin.

Iliac fascia

The iliac fascia overlies iliacus. Superiorly and laterally, it is attached to the inner aspect of the iliac crest, and medially it blends with the anterior layer of thoracolumbar fascia over quadratus lumborum and with the psoas fascia. Inferiorly and laterally, it continues into the thigh to fuse with the femoral sheath, while medially, it is attached to the periosteum of the ilium and iliopubic eminence at the pelvic brim. Both the femoral nerve and lateral femoral cutaneous nerve lie under the iliac fascia in the pelvis, where they can be targeted for image-guided nerve blocks (Hebbard et al 2011).

VISCERAL FASCIAE

The organization of the retroperitoneum is most easily conceptualized in terms of compartments (synonym: spaces) rather than fasciae (Dodds

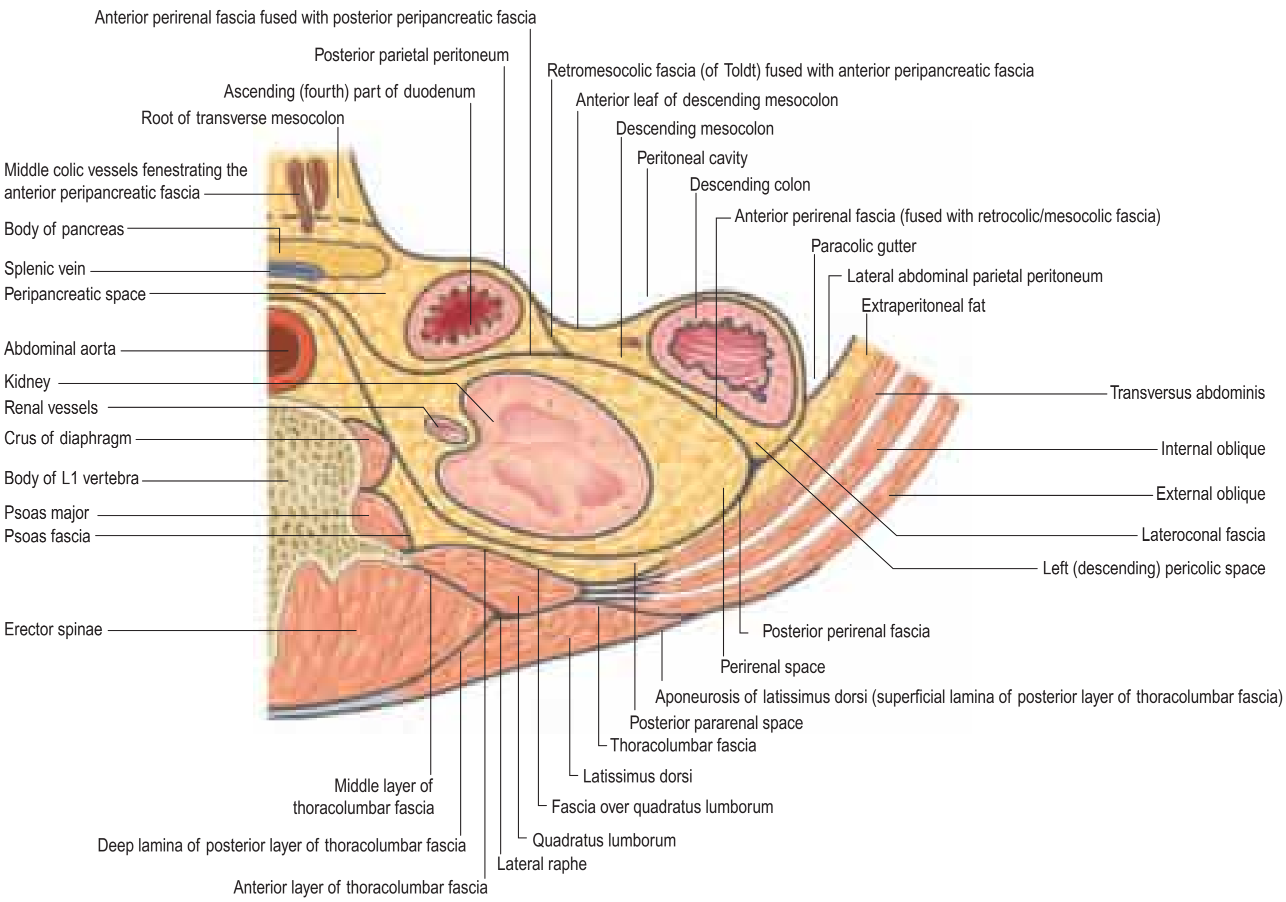

Fig. 62.1 Compartments and fasciae of the upper retroperitoneum (left side). A transverse section at approximately the level of the first lumbar vertebra, showing the location and relations of the peripancreatic space.

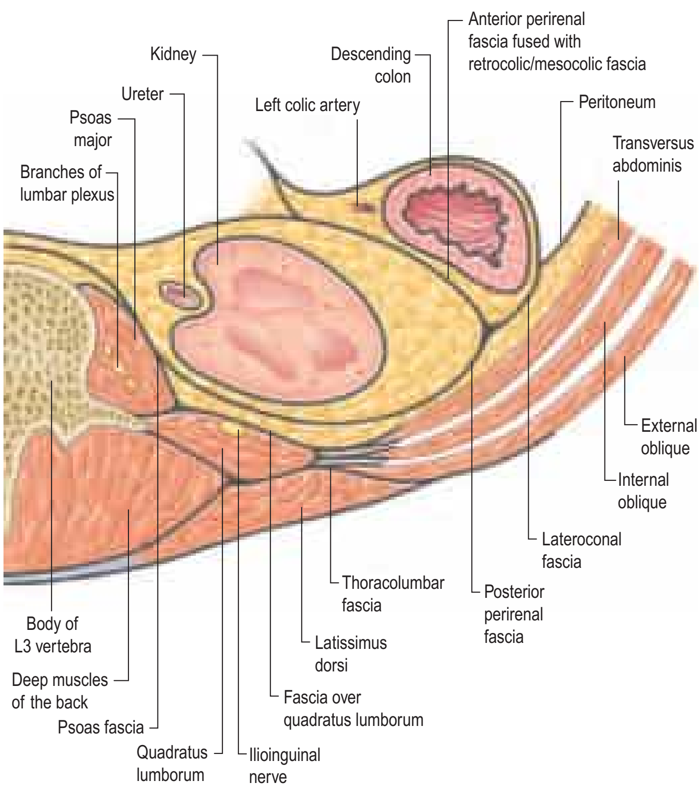

Fig. 62.2 Compartments and fasciae of the mid-retroperitoneum (left side). A transverse section at approximately the level of the third lumbar vertebra, showing the location and relations of the perirenal space.

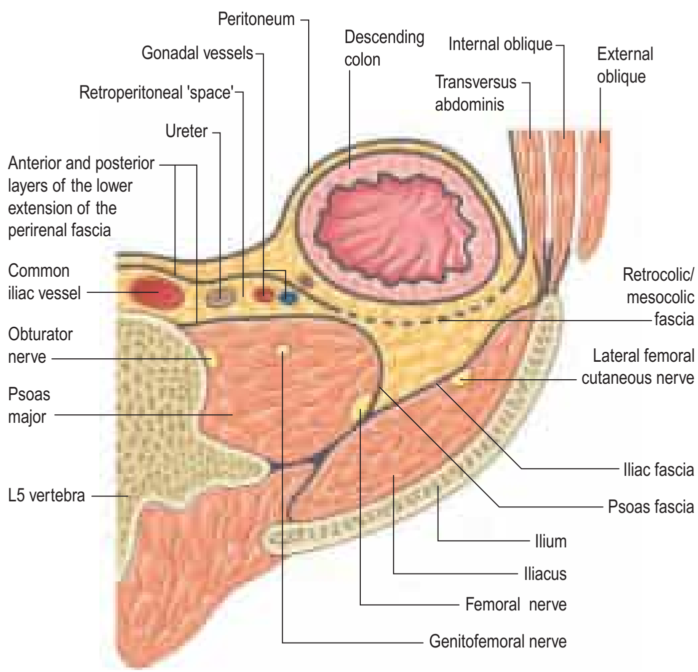

Fig. 62.3 Compartments and fasciae of the lower retroperitoneum (left side). A transverse section at approximately the L4 vertebral level, showing the location and relations of the pericolic space.

et al 1986). The fasciae bounding the compartments limit or direct the spread of blood, fluid, gas or malignant disease, but are variable and often difficult to see with cross-sectional imaging.

The fascial layers located between the posterior parietal peritoneum and the quadratus lumborum have been described in confusing eponymous terms in the literature. Different authors have used the same eponym for different fasciae; different eponyms have been used to refer to the same fascia. Thus, 'Gerota's fascia' has been used to refer to the entire perirenal fascia or to its anterior or posterior layers (Chesbrough et al 1989). 'Toldt's fascia' has been used to refer to the fusion fascia behind the ascending or descending colon (Culligan et al 2013), or to the posterior perirenal fascia, or to the fusion fascia behind the tail of the pancreas (Kimura et al 2010). 'Zuckerkandl's fascia' has been used to refer to the posterior or anterior perirenal fascia (Chesbrough et al 1989). 'Treitz's fascia' has been used to refer to the fusion fascia behind the head of the pancreas (Kimura et al 2010).

The concept of 'fusion fascia' describes a retroperitoneal fascia formed by the fusion of an embryonic mesentery with embryonic retroperitoneum. It is an avascular layer that allows a dissection plane to be developed, and also limits the spread of disease. The retropancreaticoduodenal fascia and the retrocolic/retromesocolic fascia (right and left) are fusion fascias, and delimit their respective compartments. They were first described in detail by the Austrian anatomist Carl Toldt, and only later reported in the English literature (Congdon et al 1942).

Traditional compartmental anatomy of the retroperitoneum divides it into the anterior and posterior pararenal spaces, and the perirenal space. 'Anterior pararenal space' describes the compartment between the posterior peritoneum and the anterior perirenal fascia. This is an oversimplification and is unable to explain all patterns of disease containment or spread. The smallest set of compartments sufficient to explain most such phenomena is as follows.

Perirenal space

The paired perirenal space is bounded by the perirenal fascia, which envelops the kidney and suprarenal gland on each side. Perirenal fascia is traditionally described as having anterior and posterior layers, which are continuous with each other laterally (see Figs 62.1–62.3; Ch. 74). The perirenal fascia can usually be identified on cross-sectional imaging as a thin layer surrounding the kidney and suprarenal gland, separated from the renal capsule by a variable amount of perirenal fat. Inferiorly, the perirenal space continues down around the ureter but becomes progressively narrower and may or may not extend into the pelvic retroperitoneum. Medially, the two perirenal spaces may interconnect anterior to the aorta and inferior vena cava (Kneeland et al 1987). The existence of a fascial partition separating the kidney and ipsilateral suprarenal gland within the perirenal fascial envelope is controversial (Amin et al 1976).

Anterior and posterior pararenal spaces

The anterior pararenal space (an oversimplification) refers to the region between the anterior layer of perirenal fascia and the posterior parietal peritoneum; it is not a single compartment. On the right, it contains the second part of the duodenum, ascending colon and its mesentery, and on the left, the fourth part of the duodenum, descending colon and its mesentery (see Figs 62.1–62.3).

The potential compartment between the posterior layer of perirenal fascia and the thoracolumbar and psoas fasciae on each side is the posterior pararenal space. It contains a variable amount of fat. Laterally, it is continuous with the extraperitoneal (properitoneal) fat of the anterolateral abdominal wall, and inferiorly, with the retroperitoneal fat overlying iliacus and the pelvic wall, and the perivesical fat (Coffin et al 2015).

When the ascending or descending colon is retrorenal (an anatomical variant), the colon and pericolic space lie in a groove between the posterior layer of perirenal fascia and the posterior pararenal space.

Peripancreatic space

The peripancreatic space contains the duodenum and pancreas along with the origins of the superior mesenteric vessels and the retroperitoneal segments of the common bile duct, portal vein and hepatic artery. The posterior boundary is a fusion fascia formed by fusion of the embryonic duodenal mesentery to the embryonic retroperitoneum (Dodds et al 1986). This fascia is termed 'retropancreatic' or 'retropancreaticoduodenal' fascia in the imaging literature, and 'fusion fascia of Treitz' in the surgical literature (Kimura et al 2010). The fascia forms a relatively bloodless dissection plane, used to mobilize the duodenum and head of the pancreas.

The lateral extent of the peripancreatic space is variable; it lies posterior to the ascending and descending colon and anterior to the perirenal space (see Fig. 62.1). At the tail of the pancreas, the peripancreatic space lies in continuity with the splenorenal ligament. Anteriorly, it is in continuity with the transverse mesocolon and small bowel mesentery. Consequently, peripancreatic fluid collections may extend into both mesenteries and extend posterior to the ascending and descending colon but do not usually cross into the perirenal and pericolic spaces.

Pericolic spaces

The ascending and descending pericolic spaces are narrow compartments surrounding the respective parts of the colon and in continuity medially with the mesocolon (see Figs 62.1–62.3). The pericolic spaces contain a variable amount of fat and are limited anteriorly, superiorly and laterally by the colonic serosa. Inferiorly, the pericolic spaces are continuous with the retroperitoneal spaces of the iliac fossae. Posteriorly, each pericolic space is limited by a fusion fascia formed by the right leaf of the embryonic mesocolon fusing to the left leaf of the embryonic duodenal mesentery and embryonic retroperitoneum further laterally. This retrocolic fascia (of Toldt) (Culligan et al 2013, Culligan et al 2014) forms the classic bloodless plane of dissection when performing a hemicolectomy. Its plane is entered by incising the junction of the colonic serosa with the parietal peritoneum at the white line (of Toldt) in the paracolic gutter. Radiologically, the retrocolic fascia may be indistinguishable from the anterior perirenal fascia when lateral to the duodenum and pancreas.

Variation in colonic retroperitonealization may produce a free mesocolic pedicle instead (most commonly seen at the caecum).

Lateroconal fascia

Lateroconal fascia was originally defined as the fascial layer extending from the junction of the anterior and posterior perirenal fasciae to the parietal peritoneum in the lateral paracolic gutter (Congdon and Edson 1941). It is better understood as the lateral extension of the retrocolic fusion fascia that blends with the parietal peritoneum (see Figs 62.1–62.3).

The various compartments described above are illustrated in the context of clinical disease in **Figures 62.4–62.6**.

BONES

The posterior abdominal wall is supported by the vertebral column and bony pelvis. The individual bones are the lower two ribs, the twelfth thoracic and five lumbar vertebrae, and the sacrum and ilium, together with their interconnecting ligaments (Chs 43, 53).

MUSCLES

The majority of the muscles of the posterior abdominal wall are functionally part of the lower limb or vertebral column. They provide the surface against which the neurovascular and visceral structures of the retroperitoneum lie (**Fig. 62.7**; see Fig. 62.14).

Quadratus lumborum

Quadratus lumborum is an irregularly shaped quadrilateral muscle, broader at its inferior attachment than superiorly.

Attachments The inferior attachment is by aponeurotic fibres to the iliac crest over an area 5–7 cm lateral to the tip of the L4 transverse process and/or the iliolumbar ligament. The superior attachment is to the lower anterior surface of the twelfth rib, the lateral surface of the twelfth thoracic vertebra, and the apices of the transverse processes of the upper four lumbar vertebrae. Fascicles vary in number and size but are arranged in three layers: anterior, middle and posterior (Phillips et al 2008).

Relations Anteriorly are the colon (ascending on the right, descending on the left), kidney, psoas major and minor, and diaphragm. The subcostal, iliohypogastric and ilioinguinal nerves lie on the fascia anterior to the muscle, bound down to it by the medial continuation of the transversalis fascia.

Vascular supply Quadratus lumborum is supplied by branches of the lumbar arteries, the arteria lumbales imae from the median sacral artery, the lumbar branch of the iliolumbar artery, and branches of the subcostal artery.

Innervation The muscle is innervated by the ventral rami of the twelfth thoracic and upper three or four lumbar spinal nerves.

Actions Quadratus lumborum fixes the twelfth rib, and acts as a muscle of inspiration by helping to stabilize the lower attachments of the diaphragm. With the pelvis fixed, unilateral contraction flexes the vertebral column to the same side, and bilateral contraction probably helps to extend the lumbar part of the vertebral column. These actions on the lumbar spine are reported to be weak (Phillips et al 2008) and yet the muscle undergoes considerable hypertrophy in some sporting activities (Ranson et al 2008).

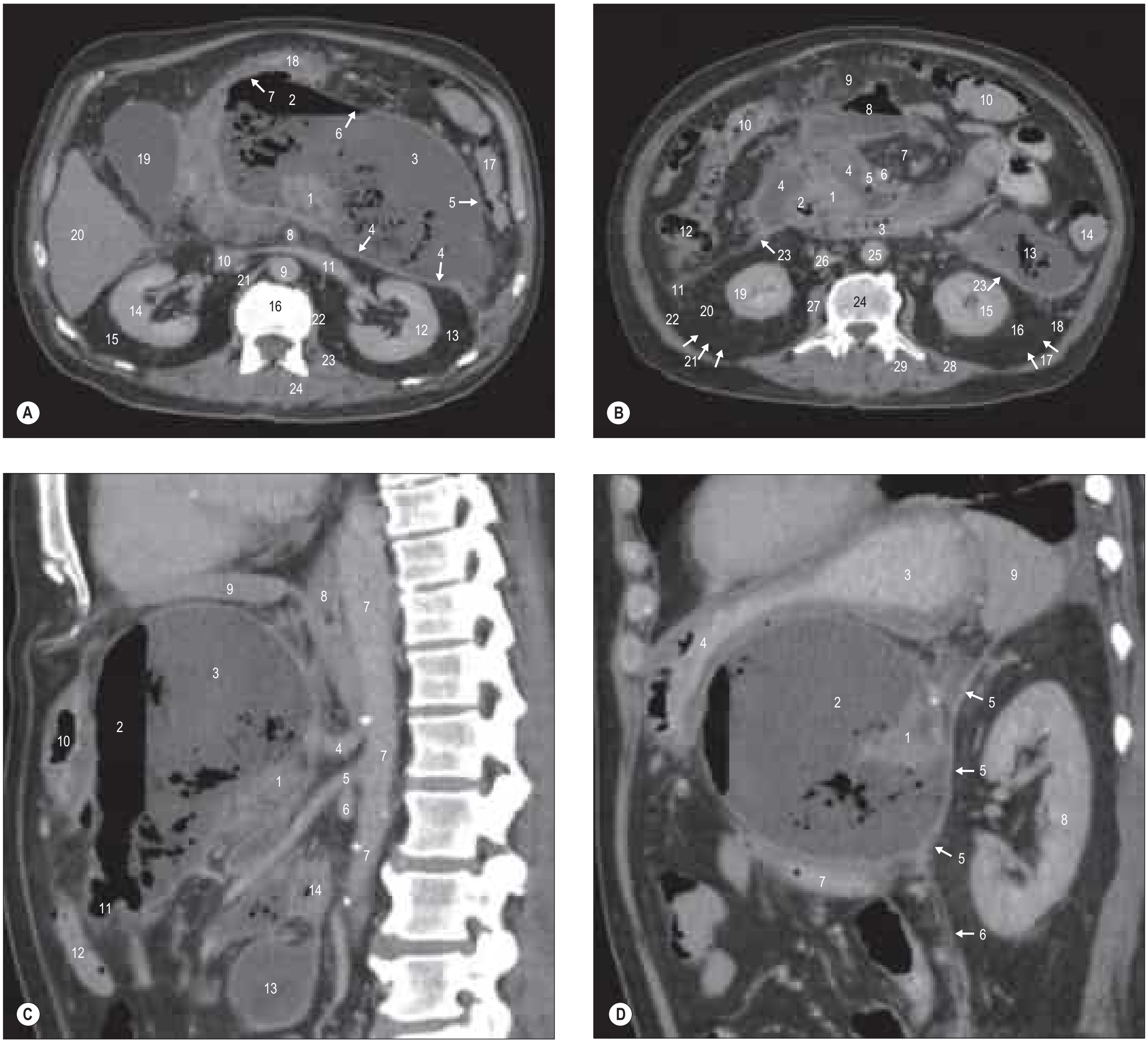

Fig. 62.4 A 58-year-old male with chronic pancreatitis and a peripancreatic pseudocyst. Oral and intravenous contrast-enhanced computed tomography (CT; axial and sagittal planes). The pseudocyst, containing gas and fluid, has expanded the peripancreatic space. The fusion fasciae forming the boundaries of the peripancreatic space are thickened through chronic inflammation and are easily visible. **A**, A transverse section at the level of the body of the second lumbar vertebra. Key: 1, neck of pancreas; 2, gas in anterior part of collection (outlining peripancreatic space); 3, fluid and debris in posterior (dependent) part of collection (outlining peripancreatic space); 4, retropancreatic fusion fascia (anterior perirenal fascia fused to right leaf of mesoduodenum); 5, anterior peripancreatic fascia fused to right leaf of left mesocolon; 6, anterior peripancreatic fascia deep to left mesocolon and collapsed lesser sac; 7, anterior peripancreatic fascia fused to peritoneum of lesser sac in gastric bed; 8, superior mesenteric artery; 9, abdominal aorta; 10, inferior vena cava; 11, left renal vein; 12, left kidney parenchyma; 13, left perirenal space (not distinguishable from left posterior pararenal space); 14, right kidney parenchyma; 15, right perirenal space (not distinguishable from right posterior pararenal space); 16, upper L2 vertebral body; 17, splenic flexure; 18, stomach (elevated and compressed by collection); 19, gallbladder; 20, right lobe of liver (segments 5 and 6); 21, right crus of diaphragm; 22, origin of left psoas major; 23, left quadratus lumborum; 24, left erector spinae group. **B**, A transverse section at the level of the body of the fourth lumbar vertebra. Key: 1, inferior part of body and uncinate process of pancreas; 2, distal second part of duodenum; 3, third part of duodenum; 4, peripancreatic space collection (around and anterior to pancreas and duodenum); 5, superior mesenteric vein; 6, superior mesenteric artery; 7, small bowel mesentery; 8, part of collection dissecting into transverse mesocolon; 9, transverse mesocolon (fused to greater omentum); 10, transverse colon; 11, collection dissecting behind ascending colon; 12, ascending colon; 13, collection (with gas) dissecting behind descending colon; 14, descending colon; 15, left kidney parenchyma; 16, left perirenal space; 17, left posterior perirenal fascia (normal thickness, therefore faint); 18, left posterior pararenal space; 19, right kidney parenchyma; 20, right perirenal space; 21, right posterior perirenal fascia; 22, right posterior pararenal space; 23, anterior pararenal fascia (thickened through chronic inflammation); 24, upper L4 vertebral body; 25, abdominal aorta; 26, inferior vena cava; 27, right psoas major; 28, left quadratus lumborum; 29, left erector spinae group. **C**, A mid-sagittal section. Key: 1, neck of pancreas; 2, gas in anterior part of collection; 3, fluid in posterior part of collection; 4, coeliac trunk; 5, superior mesenteric artery; 6, left renal vein; 7, abdominal aorta; 8, diaphragmatic crura; 9, left lobe of liver; 10, antrum of stomach (collapsed); 11, part of collection dissecting into transverse mesocolon; 12, transverse colon; 13, part of collection dissecting into small bowel mesentery; 14, third part of duodenum. **D**, A left sagittal section through left renal hilum. Key: 1, body of pancreas; 2, fluid collection; 3, fundus of stomach (containing oral contrast); 4, body of stomach (containing oral contrast and displaced); 5, left anterior perirenal fascia fused with retropancreatic fascia; 6, left anterior perirenal fascia fused with right leaf of left mesocolon (fusion fascia of Toldt); 7, fourth part of duodenum; 8, left kidney parenchyma; 9, spleen.

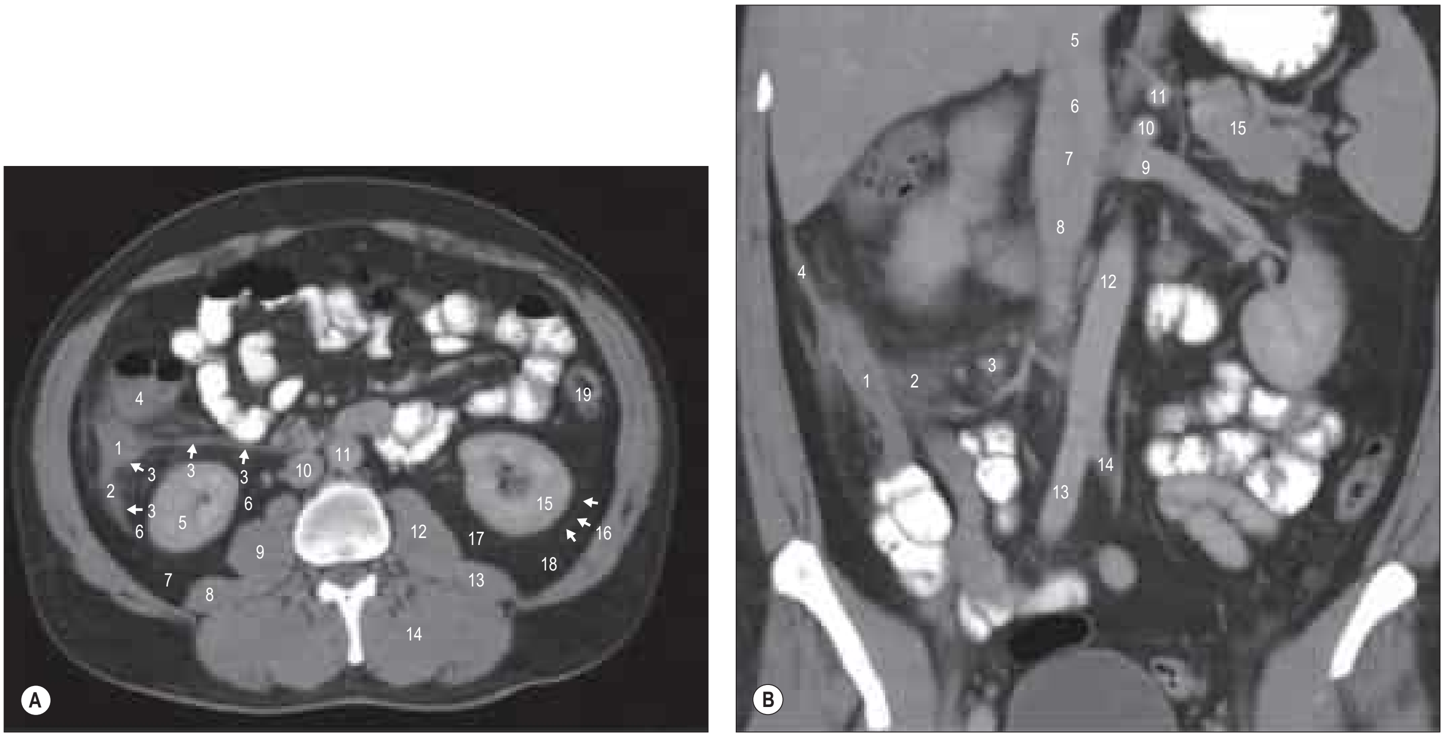

Fig. 62.5 A 43-year-old male with subacute retrocaecal appendicitis. Inflammatory fluid and stranding in the right pericolic space. The fused right anterior perirenal fascia and right mesocolon (the fusion fascia of Toldt) is visible. **A**, A transverse section through the vermiform appendix. Key: 1, thickened and inflamed retrocaecal appendix; 2, inflammatory fluid tracking posteriorly in right pericolic space, anterior to right perirenal space; 3, anterior perirenal fascia (fused with retrocolic/mesocolic) fascia; 4, caecum/ascending colon; 5, right kidney parenchyma; 6, right perirenal space 7, right posterior pararenal space (the posterior perirenal fascia is difficult to identify); 8, right quadratus lumborum; 9, right psoas major; 10, inferior vena cava; 11, abdominal aorta; 12, left psoas major; 13, left quadratus lumborum; 14, left erector spinae group; 15, left kidney parenchyma; 16, left posterior perirenal fascia; 17, left perirenal space; 18, left posterior pararenal space; 19, descending colon. **B**, A coronal section through the appendix. Key: 1, thickened and inflamed retrocaecal appendix; 2, inflammatory fluid and oedema extending along right mesocolon towards root of mesentery; 3, enlarged inflamed right ileocolic lymph node; 4, inflammatory fluid tracking in right mesocolon; 5, inferior vena cava – retrohepatic segment; 6, inferior vena cava – infrahepatic peritonealized segment; 7, inferior vena cava – renal segment; 8, inferior vena cava – infrarenal segment; 9, left renal vein; 10, superior mesenteric artery; 11, coeliac trunk; 12, abdominal aorta; 13, right common iliac artery; 14, left common iliac artery (origin); 15, tail of pancreas.

Fig. 62.6 A 61-year-old male with mid-ureteric calculus causing high-grade obstruction of the left ureter and extravasation of urine from the left renal hilum. Oral and intravenous contrast-enhanced CT. A transverse section through the left renal hilum (the calculus is below the plane of this section). Key: 1, left kidney parenchyma; 2, extravasated urine in left perirenal space; 3, posterior perirenal fascia; 4, anterior perirenal fascia (fused medially with retropancreatic fascia and fused laterally with right leaf of descending mesocolon); 5, left posterior pararenal space; 6, descending colon; 7, third part of duodenum; 8, inferior vena cava; 9, abdominal aorta; 10, right psoas major; 11, right quadratus lumborum; 12, right erector spinae group; 13, aortocaval lymph node (borderline enlarged); 14, left para-aortic lymph nodes (borderline enlarged).

Psoas major, psoas minor and iliacus

Psoas major and iliacus are functionally important in the lower limb and are described together with psoas minor in Chapter 80.

Posterior abdominal wall hernias

In the absence of a previous surgical incision, herniation through the posterior abdominal wall is rare because the muscular and fascial layers usually protect against protrusion of the posterior abdominal viscera, which are relatively immobile. However, the posterior free border of external oblique, the inferior free border of latissimus dorsi, and the iliac crest delimit the lumbar triangle (of Petit), an area of potential weakness through which a lumbar hernia may develop (Stamatiou et al 2009).

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

ABDOMINAL AORTA

The abdominal aorta begins at the aortic hiatus of the diaphragm, anterior to the twelfth thoracic vertebra (Mirjalili et al 2012a). It descends anterior to the lumbar vertebrae and bifurcates into two common iliac arteries anterior to the fourth lumbar vertebra or the L4/5 intervertebral disc, slightly to the left of the midline (Mirjalili et al 2012b, Moussallem et al 2012) (**Figs 62.8–62.10**). The angle of bifurcation is very variable (Moussallem et al 2012). The mean adult infrarenal aortic diameter measured by computed tomography is 19–21 mm in men and 16–18 mm in women (Rogers et al 2013), but there are ethnic variations (Jasper et al 2014). Measured by ultrasound, equivalent values are 20 mm (SD 2.5 mm) in men and 17 mm (SD 1.5 mm) in women (Needleman 2006). The mean calibre of the abdominal aorta decreases slightly from proximal to distal. With advancing age, there is a progressive increase in abdominal aortic diameter in both sexes and

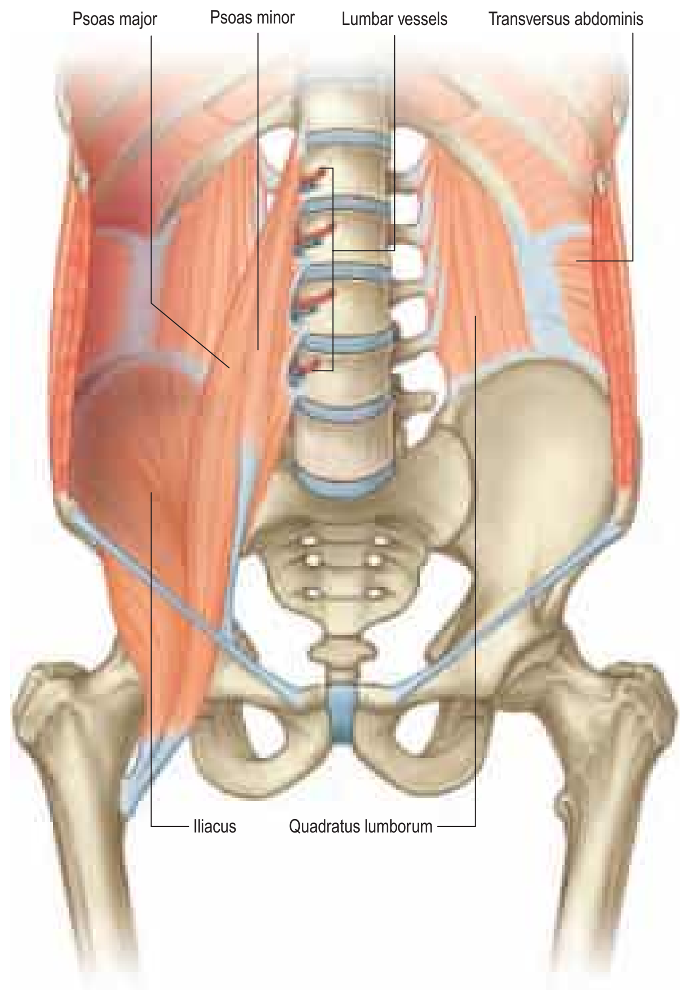

Fig. 62.7 Bones and deep muscles of the posterior abdominal wall. Left psoas major and the diaphragm have been removed. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

the tapering becomes more pronounced (Fleischmann et al 2001). In the elderly, the abdominal aorta frequently becomes ectatic and tortuous, changing the angle and position of the bifurcation and rotating the origins of the major branches.

Approximately 80% of abdominal aortic aneurysms occur in the infrarenal segment of the aorta. Men with atherosclerosis are most at risk of developing the condition, particularly from the sixth decade onwards (Takayama and Yamanouchi 2013). The high mortality risk from spontaneous rupture has prompted the development of ultrasound screening programmes to detect occult aneurysms and repair them electively. The most common clinical cut-off diameter for the presence of an abdominal aortic aneurysm is 3 cm (Needleman 2006, Wegener 1992, Allan 2006, Moeller and Reif 2000). Repair is most commonly performed by open surgery or endovascular stenting (endovascular aneurysm repair).

Relations

The upper abdominal aorta is directly related anteriorly to the coeliac trunk and its branches, autonomic nerve plexuses and lymphatics. Below the coeliac trunk, the superior mesenteric artery leaves the aorta, descending anterior to the left renal vein. Anterior to these vessels lies the body of the pancreas, with the splenic vein on its posterior surface, extending obliquely up and to the left (see Fig. 62.8). Further anteriorly, the lesser sac separates the upper abdominal aorta from the lesser omentum, stomach and left lobe of the liver. Below the pancreas, the horizontal third part of the duodenum crosses the aorta anteriorly. The most inferior part of the abdominal aorta is covered by the posterior parietal peritoneum and crossed obliquely by the origin of the small bowel mesentery.

The twelfth thoracic vertebra, the upper four lumbar vertebrae, intervening intervertebral discs and the anterior longitudinal ligament lie posterior to the abdominal aorta. Lumbar arteries arise from its dorsal aspect, and the third and fourth (and sometimes the second) left lumbar veins cross behind it to reach the inferior vena cava. The abdominal aorta may overlap the medial border of the left psoas major muscle.

On the right, the abdominal aorta is related superiorly to the cisterna chyli and thoracic duct, the azygos vein and the right crus of the diaphragm, which overlaps and separates it from the inferior vena cava and right coeliac ganglion. Below the second lumbar vertebra, it is closely applied to the left side of the inferior vena cava. This close relationship allows the development of an aortocaval fistula, which is a rare complication of aneurysmal disease or trauma. On the left, the aorta is related superiorly to the left crus of the diaphragm and left coeliac ganglion. Level with the second lumbar vertebra, it is related to the fourth part of the duodenum, the left sympathetic trunk, and the inferior mesenteric vein.

Branches

The branches of the aorta are described as anterior, lateral and dorsal (see Figs 62.8–62.10). The anterior (unpaired) and lateral (paired) branches are distributed to the viscera, while the dorsal branches supply the body wall, vertebral column, vertebral canal and its contents. The aorta terminates by dividing into the right and left common iliac arteries.

Anterior group

Coeliac trunk

The coeliac trunk is the first anterior branch and arises just below the aortic hiatus, usually at the level of the vertebral body of T12. It is 1–3 cm long and passes almost horizontally forwards and slightly to the right above the body of the pancreas and splenic vein. In most individuals, it trifurcates into the left gastric, common hepatic and splenic arteries. Variations occasionally occur and include a separate origin of the left gastric artery from the abdominal aorta, one or both inferior phrenic arteries arising from the coeliac trunk, and the superior mesenteric artery or one or more of its branches arising in common with the coeliac trunk (Panagouli et al 2013). Anterior to the coeliac trunk lies the lesser sac. The coeliac plexus surrounds the trunk, sending extensions along its branches. On the right lie the right coeliac ganglion, right crus of the diaphragm and the caudate lobe of the liver. To the left lies the left coeliac ganglion, left crus of the diaphragm and the cardiac region of the stomach. Rarely, the coeliac trunk can be compressed by the median arcuate ligament, resulting in visceral ischaemia and abdominal pain (Loukas et al 2007a).

Superior mesenteric artery

The superior mesenteric artery originates from the aorta approximately 1–2 cm below the coeliac trunk, at the level of the L1 vertebral body (Mirjalili et al 2012b) (Chs 65–66). It lies posterior to the body of the pancreas and splenic vein, and is separated from the aorta by the left renal vein. It passes forwards and inferiorly, anterior to the uncinate process of the pancreas and the third part of the duodenum, to enter the root of the small bowel mesentery and supply the midgut. Numerous anatomical variants have been described (Bergman et al 2014), the most common being an accessory or replaced right hepatic artery arising near the origin of the superior mesenteric artery. Present in about 15% of individuals, the accessory or replaced right hepatic artery courses posterior to the portal vein and ascends posterolateral to the common bile duct (Winston et al 2007).

Inferior mesenteric artery

The inferior mesenteric artery is smaller than the superior mesenteric artery. It arises from the anterior or left anterolateral aspect of the aorta at about the level of the L3 vertebral body, 3 or 4 cm above the aortic bifurcation and posterior to the inferior border of the horizontal part of the duodenum (Ch. 66).

Lateral group

Suprarenal artery

The right and left middle suprarenal arteries arise from each side of the abdominal aorta, level with the superior mesenteric artery. Each passes laterally over the crus of the diaphragm to the suprarenal gland, where it anastomoses with the suprarenal branches of the ipsilateral inferior phrenic and renal arteries (Toni et al 1988) (Ch. 71). The right middle suprarenal artery passes behind the inferior vena cava, near the right coeliac ganglion. The left middle suprarenal artery passes close to the left coeliac ganglion, splenic artery and the superior border of the pancreas.

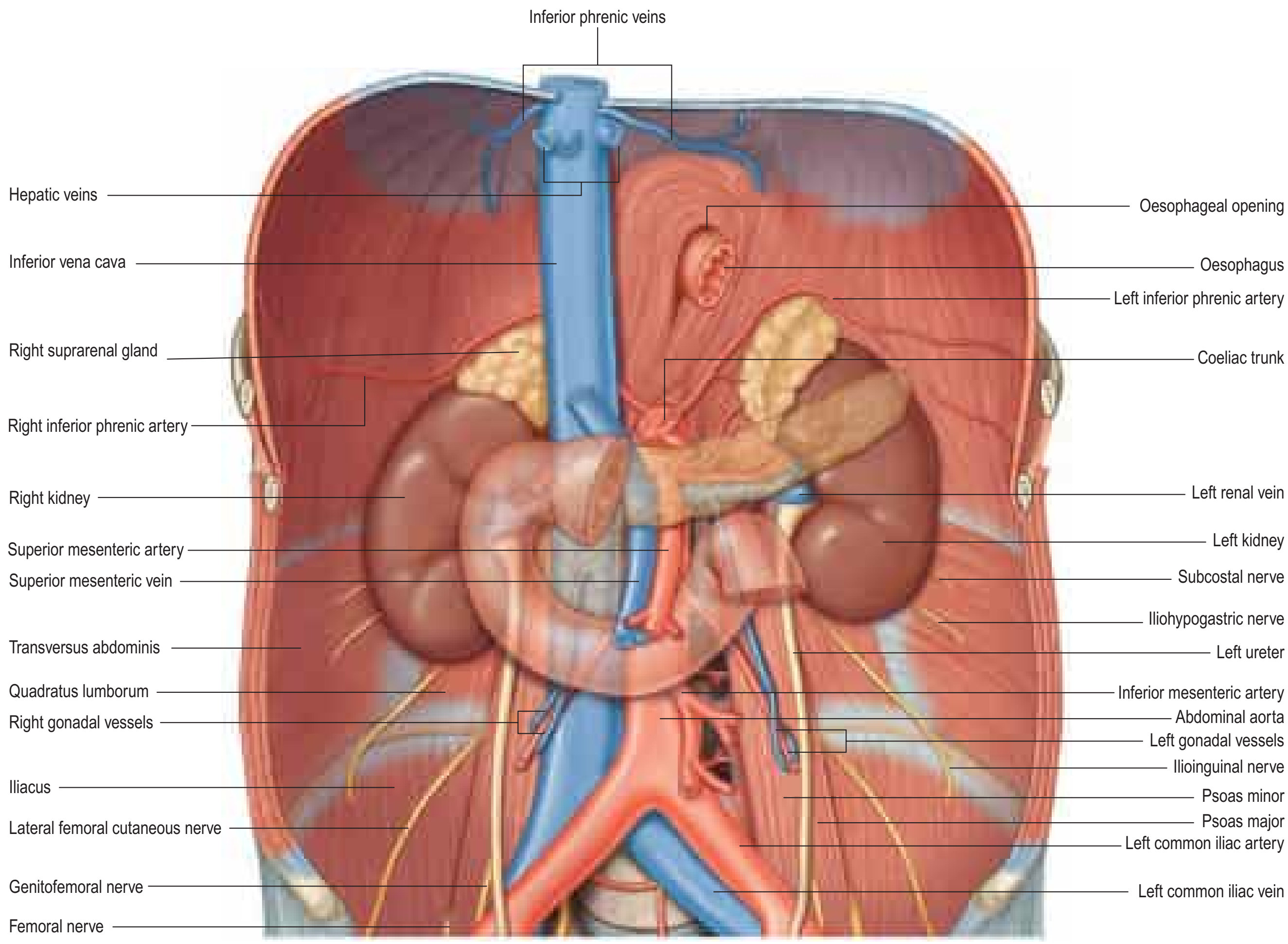

Fig. 62.8 The abdominal aorta and inferior vena cava, and their branches. The fasciae, lymphatics and connective tissue have been removed for clarity.

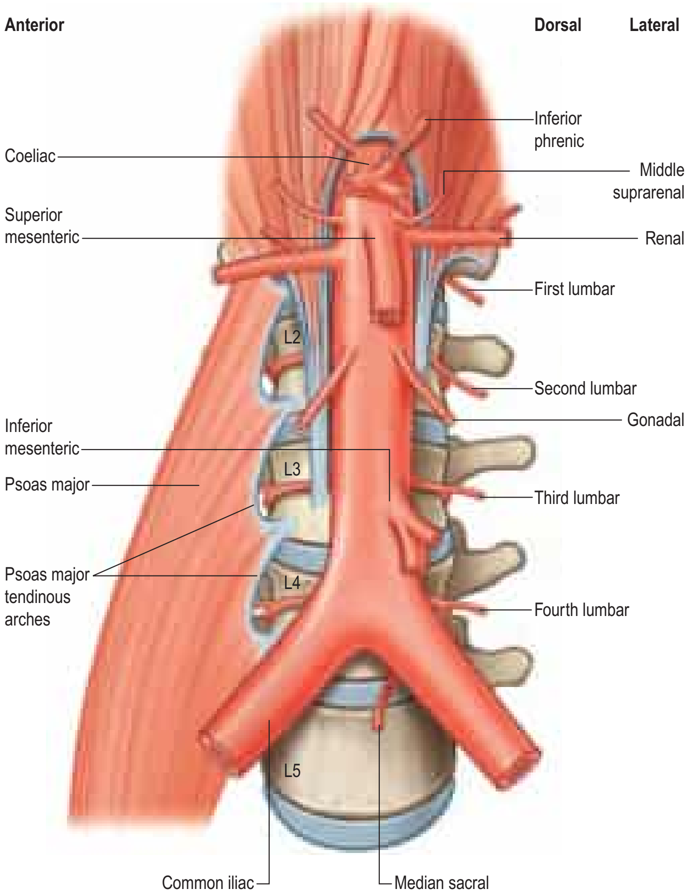

Fig. 62.9 The branches of the abdominal aorta.

Renal artery

The renal arteries are two of the largest branches of the abdominal aorta and arise laterally just below the origin of the superior mesenteric artery at about the level of the L1 vertebral body (Mirjalili et al 2012b) (Ch. 74). When the arteries arise at different cranio-caudal levels, the right ostium is more commonly higher than the left. The right renal artery is longer and passes posterior to the inferior vena cava, right renal vein, head of the pancreas and second part of the duodenum. The left renal artery passes behind the left renal vein, the body of the pancreas and the splenic vein. Variations in the number, origin, course and branching patterns of the renal arteries are common.

Gonadal artery

The gonadal arteries are two long, slender vessels that arise from the aorta a little inferior to the renal arteries. Each passes inferolaterally under the parietal peritoneum on psoas major to supply the ipsilateral gonad (Chs 76–77). One or both gonadal arteries may arise from a renal artery or be double.

Dorsal group

Inferior phrenic arteries

The inferior phrenic arteries usually arise either from the aorta, just above the level of the coeliac trunk, or directly from the coeliac trunk; occasionally, they originate from the renal artery (Loukas et al 2005a, Gwon et al 2007). They contribute to the arterial supply of the diaphragm. Each artery ascends laterally, anterior to the crus of the diaphragm, near the medial border of the ipsilateral suprarenal gland. It then divides into an ascending and a descending branch. The left ascending branch passes behind the oesophagus and then runs anteriorly on the left side of the oesophageal hiatus, where it bifurcates; one branch curves forwards to anastomose with its counterpart in front of the central tendon of the diaphragm and the other branch approaches the thoracic wall to anastomose with the musculophrenic and pericardiacophrenic arteries. The right ascending branch passes behind the inferior vena cava and then bifurcates; one branch runs anteriorly on the right side of the diaphragmatic opening for the inferior vena cava (which it supplies) before anastomosing with its counterpart in front of the central tendon of the diaphragm, and the other branch passes

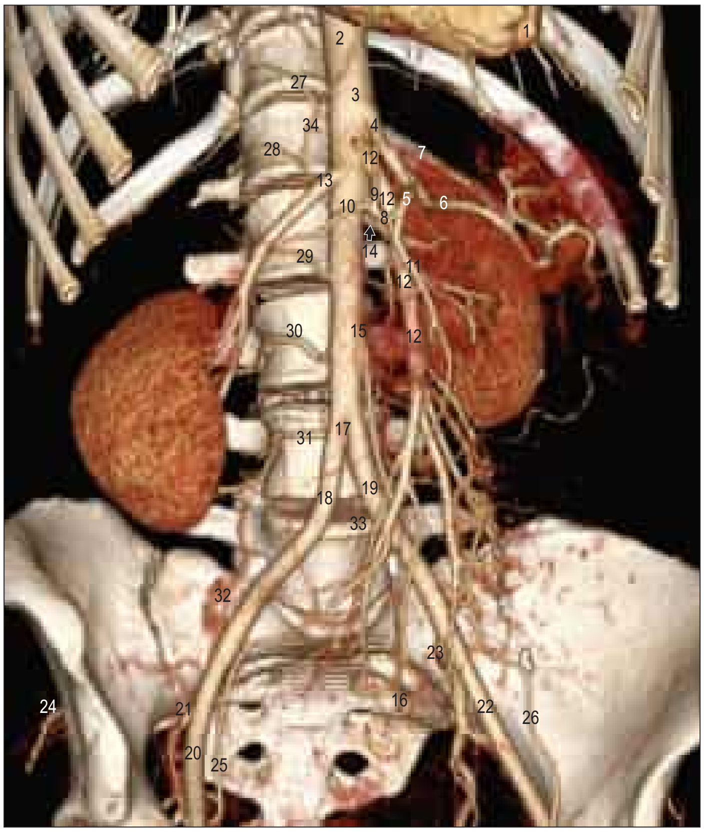

Fig. 62.10 The abdominal aorta and its branches. A 36-year-old female investigated by CT aortogram for possible Marfan's aortic dissection; only the normal abdominal aorta part of the dataset is shown in this three-dimensional reconstruction. Surface shaded display. Key: 1, apex of left ventricle; 2, low thoracic aorta; 3, approximate position of diaphragmatic hiatus; 4, coeliac trunk; 5, common hepatic artery; 6, splenic artery; 7, left gastric artery; 8, proper hepatic artery; 9, left hepatic artery; 10, right hepatic artery; 11, gastroduodenal artery; 12, superior mesenteric artery; 13, right renal artery; 14, left renal artery partly obscured by proper hepatic artery lying more superficial (with arrow); 15, inferior mesenteric artery; 16, superior rectal artery; 17, aortic bifurcation in front of L4; 18, right common iliac artery; 19, left common iliac artery; 20, right external iliac artery; 21, right internal iliac artery; 22, left external iliac artery; 23, left internal iliac artery; 24, right superior gluteal artery; 25, right inferior epigastric artery; 26, left inferior epigastric artery; 27, right T12 segmental artery; 28, right L1 segmental artery; 29, right L2 segmental artery; 30, right L3 segmental artery; 31, right L4 segmental artery; 32, right iliolumbar artery; 33, median sacral artery; 34, right superior epigastric artery (continuing from right internal thoracic artery).

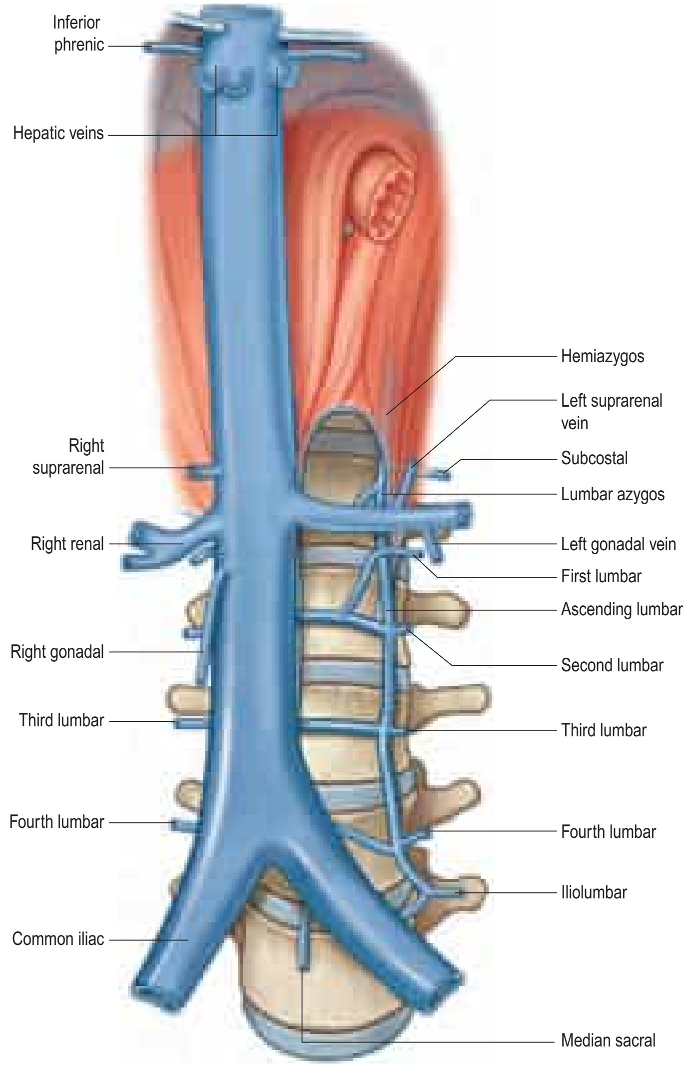

Fig. 62.11 Tributaries of the inferior vena cava and lumbar veins. Only the left lumbar venous system is shown, for clarity.

laterally on the undersurface of the diaphragm. The descending branches on each side supply the muscular diaphragm and anastomose with the lower posterior intercostal and musculophrenic arteries. Each inferior phrenic artery gives off two or three small suprarenal branches. The abdominal oesophagus, capsule of the liver, and upper pole of the spleen may also receive small arterial twigs.

The inferior phrenic artery may be a source of significant collateral blood flow to large hepatocellular cancers and is sometimes specifically occluded, along with the relevant hepatic artery, when treating such tumours by arterial embolization.

Lumbar arteries

There are usually four lumbar arteries on each side, in series with the posterior intercostal arteries. They arise from the posterolateral aspect of the abdominal aorta, opposite the lumbar vertebrae. A fifth, smaller, pair occasionally arise from the median sacral artery, but lumbar branches of the iliolumbar arteries often take their place. The lumbar arteries run posterolaterally on the first to the fourth lumbar vertebral bodies, passing behind the sympathetic trunk and tendinous arches formed by the attachments of psoas major to the vertebral bodies. The right lumbar arteries pass posterior to the inferior vena cava. The upper two right lumbar arteries and the first left lumbar artery lie behind the corresponding crus of the diaphragm. Just beyond the intervertebral foramina, each lumbar artery divides into a medial branch, which gives off spinal and ganglionic branches; a middle branch, from which dorsal and anastomotic branches arise; and a lateral branch, which supplies the abdominal wall (Arslan et al 2011). Of particular importance is the spinal branch known as the arteria radicularis magna (the artery of Adamkiewicz) (p. 770), which frequently originates from an upper lumbar artery, particularly on the left side (Biglioli et al 2004) (Ch. 45). Injury to this vessel, e.g. during thoracoabdominal aortic surgery, can cause spinal cord infarction.

The lateral branch of each lumbar artery runs posterior to psoas major and the lumbar plexus, then across the anterior surface of quadratus lumborum, before piercing the posterior limit of transversus abdominis to run forwards between it and the internal oblique. Perforating branches pass posteriorly to supply the muscles and skin of the posterior abdominal wall (Kiil et al 2009). The lumbar arteries anastomose with one another and the lower posterior intercostal, subcostal, iliolumbar, deep circumflex iliac and inferior epigastric arteries.

The dorsal branch of each lumbar artery passes backwards between the adjacent transverse vertebral processes to supply the dorsal muscles, vertebrae, joints and skin of the back.

Median sacral artery

The median sacral artery is a small branch that arises from the posterior aspect of the aorta a little above its bifurcation. It descends close to the midline, anterior to the fourth and fifth lumbar vertebrae, sacrum and coccyx. At the level of the fifth lumbar vertebra, it is crossed by the left common iliac vein and often gives off a small lumbar artery (arteria lumbales imae), small branches of which reach the anorectum via the anococcygeal ligament. Anterior to the fifth lumbar vertebra, the median sacral artery anastomoses with a lumbar branch of the iliolumbar artery. Anterior to the sacrum, it anastomoses with the lateral sacral arteries and sends branches into the anterior sacral foramina.

INFERIOR VENA CAVA

The inferior vena cava (**Figs 62.11–62.12**) conveys blood to the right atrium from almost all of the structures below the diaphragm. Most of

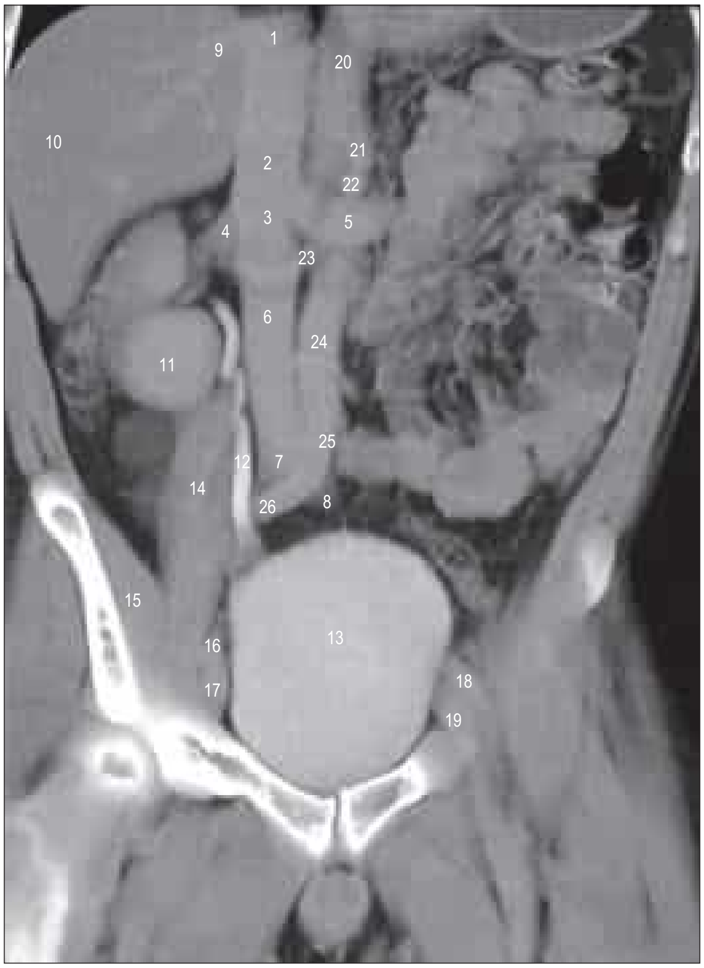

Fig. 62.12 A 41-year-old male investigated for microscopic haematuria. CT intravenous urogram. Normal study. Late phase acquisition. Oblique coronal, mid-abdominal 15 mm slab, maximal intensity projection (MIP) reformats. Key: 1, inferior vena cava, hepatic segment; 2, inferior vena cava, peritonealized segment; 3, inferior vena cava, renal segment; 4, right renal vein; 5, left renal vein crossing in front of the aorta; 6, inferior vena cava, infrarenal segment; 7, inferior vena cava, confluence (partly obscured by right common iliac artery); 8, left common iliac vein (coursing posteriorly out of slab; right common iliac vein obscured); 9, right hepatic vein joining inferior vena cava; 10, liver parenchyma; 11, right kidney lower pole parenchyma; 12, right ureter (with excreted contrast); 13, urinary bladder (with excreted contrast); 14, right psoas major; 15, right iliacus; 16, right external iliac artery; 17, right external iliac vein; 18, left external iliac artery; 19, left external iliac vein; 20, low thoracic aorta; 21, coeliac axis; 22, superior mesenteric artery; 23, right renal artery passing behind inferior vena cava; 24, aorta, infrarenal segment; 25, aortic bifurcation (partly obscured); 26, right common iliac artery (left common iliac artery courses posteriorly out of slab).

its course is within the abdomen, but a small segment lies within the pericardium in the thorax.

The inferior vena cava is formed by the junction of the left and right common iliac veins anterior to the fifth lumbar vertebral body, about 1 cm to the right of the midline. It ascends anterior to the vertebral column, to the right of the aorta, and lies in a deep groove on the posterior surface of the liver, sometimes completely embedded by liver tissue. It traverses the central tendon of the diaphragm between its median and right 'leaves', and passes through the fibrous pericardium and a posterior inflection of the serous pericardium to open into the posteroinferior part of the right atrium. The abdominal portion of the inferior vena cava is devoid of valves.

The inferior vena cava can be encircled and controlled between the renal veins below and the hepatic veins above (e.g. when operating on a renal cell carcinoma extending into the inferior vena cava).

Relations of the abdominal part of the inferior vena cava

The inferior vena cava lies behind the peritoneum of the posterior abdominal wall. At its origin, it is related anteriorly to the right common iliac artery. From below upwards, its anterior surface is crossed obliquely by the root of the small bowel mesentery and its contained vessels and nerves, the right gonadal artery and the third part of the duodenum. Further cranially, it lies behind the head of the pancreas and first part of the duodenum, separated from these structures by the common bile duct and portal vein. Above the duodenum, its anterior surface is covered by the peritoneum of the posterior abdominal wall, which forms the posterior wall of the epiploic foramen, and which separates the inferior vena cava from the right free border of the lesser omentum and its contents. Above this, it is intimately related to the liver anteriorly.

The posterior relations of the inferior vena cava include the lower three lumbar vertebral bodies and their intervertebral discs, the anterior longitudinal ligament, sympathetic trunk, right third and fourth lumbar arteries, and the right psoas major. Superior to these structures, the inferior vena cava is related posteriorly to the right renal and middle suprarenal arteries, the medial part of the right suprarenal gland, the right coeliac ganglion and the right inferior phrenic arteries.

The right ureter, medial border of the right kidney, second part of the duodenum, and the right lobe of the liver are all lateral to the right side of the inferior vena cava. The abdominal aorta, right crus of the diaphragm and the caudate lobe of the liver are left-sided relations.

The normal diameter of the adult inferior vena cava is up to 30 mm (Moeller and Reif 2000); its cross-sectional shape and calibre reflect the degree of venous filling. Anatomical variants of the inferior vena cava related to its complex embryogenesis are well described. Among these are a double inferior vena cava (the left-sided vessel usually joins the left renal vein); azygos continuation of the inferior vena cava; or a left-sided inferior vena cava (which may exist in isolation or as part of situs inversus) (Ang et al 2013, Spentzouris et al 2014).

Tributaries

The inferior vena cava usually receives the common iliac veins at its origin and the lumbar, right gonadal, renal, right suprarenal, hepatic and inferior phrenic veins throughout its course (see Figs 62.11–62.12).

Lumbar veins

Four pairs of lumbar veins collect blood from the territories supplied by the corresponding lumbar arteries (see above), including the dorsal, lateral and anterior abdominal wall. These branches anastomose posteriorly with tributaries of the azygos and hemiazygos veins, and anteriorly with branches of the epigastric, circumflex iliac and lateral thoracic veins. These superficial anastomoses provide alternative routes of venous drainage from the pelvis and lower limbs to the heart in the presence of inferior vena caval obstruction. The lumbar veins also communicate with the external and internal vertebral venous plexuses, providing an additional collateral pathway for venous return. The paired longitudinal ascending lumbar veins connect ipsilateral lumbar veins. The third and fourth lumbar veins usually pass forwards on the sides of the corresponding vertebral bodies to enter the posterior aspect of the inferior vena cava; the left lumbar veins pass behind the abdominal aorta and are, therefore, longer. The first and second lumbar veins are much more variable; they may drain into the inferior vena cava, ascending lumbar, lumbar azygos and renal vein (on the left), and are often connected to each other. Indeed, the first lumbar vein often passes inferiorly to join the second lumbar vein or, less commonly, drains directly into the ascending lumbar vein or passes forwards over the L1 vertebral body to join the lumbar azygos vein. The second lumbar vein may join the inferior vena cava at or near the level of the renal veins or, less commonly, joins the third lumbar or ascending lumbar vein.

Ascending lumbar veins

The paired ascending lumbar veins run medial to psoas major and connect ipsilateral common iliac, iliolumbar and lumbar veins. The ascending lumbar vein is variable in its course and connections; rarely, the whole vein or a segment may be absent on one side (Lolis et al 2011). It commonly joins the subcostal vein to form the azygos vein on the right and the hemiazygos on the left. The azygos and hemiazygos veins run forwards over the twelfth thoracic vertebral body, and pass deep to or through the right and left crus of the diaphragm, respectively, into the thorax (Ch. 56). The ascending lumbar vein is usually joined by a small vein, the lumbar azygos vein, from the back of the inferior vena cava or left renal vein. Sometimes, the ascending lumbar vein ends in the first lumbar vein, which then joins the lumbar azygos vein at the level of the first lumbar vertebra. Blood flow in the ascending lumbar veins can occur in either direction (Morita et al 2007).

Gonadal veins

The right gonadal vein enters the inferior vena cava directly on its right anterolateral aspect at an acute angle about 2 cm inferior to the left renal vein in adults; occasionally, it drains into the right renal vein (Barber et al 2012). The left gonadal vein terminates in the left renal vein. Both veins may be replaced by multiple vessels in the lower abdomen and sometimes remain double as far as their termination. Of the two gonadal veins, the left is more prone to develop venous incompetence.

Renal veins

The renal veins are large-calibre vessels, which lie anterior to the renal arteries and open into the inferior vena cava almost at right angles (Ch. 74). The left is three times longer than the right (approximately 7.5 cm and 2.5 cm, respectively). The left vein lies on the posterior abdominal wall posterior to the splenic vein and body of the pancreas. It passes to the right in the angle between the abdominal aorta posteriorly and the superior mesenteric artery anteriorly, to empty into the inferior vena cava. Occasionally, the left renal vein or an accessory left renal vein is retro-aortic. The right renal vein lies posterior to the second part of the duodenum and, sometimes, the lateral part of the head of the pancreas.

Suprarenal vein

Most commonly, a single vein drains each suprarenal gland (Ch. 71). The short right suprarenal vein drains directly into the inferior vena cava at the level of the twelfth thoracic vertebra; the longer left vein usually joins the left renal vein and may receive the left inferior phrenic vein. Numerous anatomical variations have been described (Cesmebasi et al 2014).

Inferior phrenic veins

The inferior phrenic veins usually originate on the superior surface of the diaphragm but run mostly on its inferior surface to drain into the posterolateral aspect of the inferior vena cava. Both veins receive several diaphragmatic tributaries. The left vein runs to the left of the oesophageal hiatus, whereas the right courses to the right of the opening in the diaphragm for the inferior vena cava, each receiving an oesophageal tributary (Loukas et al 2005b). Further, the left inferior phrenic vein frequently communicates with the left gastric vein and hence may become prominent in portal hypertension. Both inferior phrenic veins usually drain directly into the inferior vena cava just below the diaphragm, sometimes joining anteriorly to form a short common trunk. The right may occasionally drain into the right hepatic vein or inferior vena cava above the diaphragm. In contrast, the left frequently drains into the left suprarenal, left renal or left hepatic vein.

Collaterals in inferior vena caval occlusion

Obstruction of the inferior vena cava from thrombosis, embolism, extrinsic compression or intrinsic disease results in the development of an extensive venous collateral circulation via tributaries that ultimately connect to the superior vena cava. These include the azygos–hemiazygos venous system, the vertebral venous plexuses, and superficial body wall veins. The lumbar veins contribute significantly to these pathways (Golub et al 1992).

LYMPHATIC DRAINAGE

Lymph from the skin and subcutaneous tissues of the abdominal wall drains via small-calibre superficial lymphatics to axillary and inguinal nodes (Tourani et al 2013). The midline and the level of the umbilicus form inconstant and variable watershed boundaries for these drainage territories. Lymphatics from the deeper layers of the body wall and the abdominal and pelvic viscera drain almost exclusively to the cisterna chyli and thoracic duct. The former drains via ipsilateral retroperitoneal lymph nodes that are concentrated around the external iliac and common iliac vessels and along the lateral aspects of the aorta and inferior vena cava. There is considerable overlap between the lymphatic drainage basins of individual viscera. At certain sites, there is a cranio-caudal sequence of lymph nodes, e.g. para-oesophageal, retrocrural and para-aortic nodes; mediastinal and gastro-oesophageal nodes; and parasternal, superior and inferior epigastric nodes. Some lymphatic drainage occurs directly across the diaphragm to the chest from the bare area of the liver.

The paired retroperitoneal viscera drain to lateral aortic (also termed para-aortic) nodes around the origin of their arterial supply. Thus, the kidneys and suprarenal glands drain to nodes around the renal hilum and to lateral aortic nodes around the origin of the renal arteries (L1–2 vertebral level). The testes drain to para-aortic and paracaval nodes (approximately L2); the pattern of lymphatic spread from a testicular malignancy suggests that lymph from the left testis drains to left para-aortic nodes en route to the cisterna chyli, whereas lymph from the right testis drains to right and then left para-aortic nodes before reaching the cisterna chyli. Lymph from the ovary may also drain via lymphatics in the broad ligament and round ligament to internal iliac and inguinal lymph nodes, respectively. Lymphatic drainage of the ureters is less well defined but follows a regional pattern to nearby nodes.

The lymphatic drainage of the retroperitoneal colon and rectum is described in Chapter 66. The pancreas drains to numerous peripancreatic nodes (Ch. 69) as well as portal, splenic, paracaval, superior mesenteric and coeliac nodes. The lymphatic drainage of the duodenum is not well described but the lymphatics generally drain to nodal stations related to local arteries (Ch. 65). Whilst the clinical pattern of nodal metastases usually reflects normal lymphatic drainage pathways, lymphatic obstruction may lead to metastases at unusual nodal sites.

Cisterna chyli and abdominal lymph trunks

The abdominal origin of the thoracic duct usually lies to the right of the midline at the level of the twelfth thoracic vertebral body or the thoracolumbar intervertebral disc. It receives almost all the lymph from below the diaphragm via the cisterna chyli, a localized lymphatic dilation formed most commonly by the union of the intestinal lymph trunk and the left lumbar trunk (Phang et al 2014). The cisterna chyli is a saccular or fusiform lymphatic dilation measuring, on average, about 1 cm wide and 2 cm long in the cadaver (Loukas et al 2007b). Recent studies using magnetic resonance imaging (MRI) *in vivo* have shown remarkable postural variations in the cross-sectional area of the cisterna chyli, which expands considerably when moving from supine to sitting, and again from sitting to standing (Niggemann et al 2010). The cisterna chyli usually lies in front of the first and second lumbar vertebrae behind the right crus of the diaphragm to the right of the abdominal aorta but may be located at a higher vertebral level. The formation of the cisterna chyli is variable. The most common configuration, found in approximately two-thirds of individuals, is a single structure formed from the union of the intestinal lymph trunk and the left (or, less commonly, the right) lumbar lymph trunk (Loukas et al 2007b). In most other individuals, it is formed by the confluence of the intestinal and both lumbar lymph trunks, with additional small branches from intercostal lymphatics and retro-aortic lymph nodes. The upper two right lumbar arteries and the right lumbar azygos vein lie between the cisterna chyli and the vertebral column. The medial edge of the right crus of the diaphragm lies anterior to the abdominal confluence of lymph trunks.

The lumbar lymph trunks are formed by lymphatic vessels draining para-aortic nodes. The intestinal lymph trunk (which may be double) receives vessels draining from pre-aortic nodes, i.e. coeliac, superior and inferior mesenteric nodes, which drain the entire abdominal gastrointestinal tract down to the anal canal. The thoracic duct leaves the superior end of the cisterna chyli and passes through the aortic hiatus of the diaphragm posterolateral to the right side of the aorta. Its thoracic continuation is described on page 979.

The major lymphatic vessels adjacent to the abdominal aorta are at risk of injury during surgical procedures that involve peri-aortic dissection and/or lymphadenectomy. Damage to these lymphatic trunks or their obstruction by malignant disease may cause major lymph leaks manifesting as chylous ascites or chylothorax.

Retroperitoneal lymph node groups

The lymphatic drainage of the rectum, colon, stomach, pancreas, oesophagus and other organs is often described in terms of lymph node stations and levels of dissection that relate to the management of malignant disease. The terminology and classification of retroperitoneal lymph nodes are based on their location. In classic anatomical descriptions, the lymph nodes of the retroperitoneum around the abdominal aorta are grouped into pre-aortic, lateral or para-aortic, and retro-aortic groups (**Fig. 62.13**). However, it should be noted that adjacent nodal groups merge into one another with no clear demarcating boundaries. Normal lymph node dimensions are very variable. Cross-sectional imaging frequently uses 10 mm as an approximate measure for the upper limit of normal lymph node dimensions in the adult (Moeller and Reif 2000), even though some normal retroperitoneal nodes, such as the portacaval node, are typically larger.

Pre-aortic nodes

The pre-aortic nodes drain the gastrointestinal viscera, including the pancreas, liver, and spleen. These nodes lie anterior to the aorta

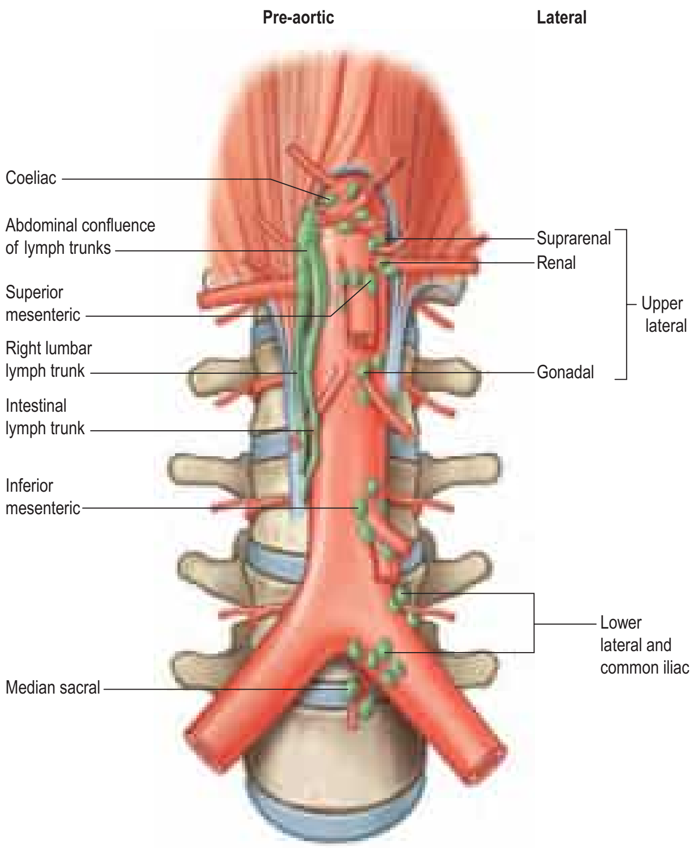

Fig. 62.13 Peri-aortic lymph node groups. The main pre-aortic groups are shown. Only the left-sided lateral aortic (para-aortic) nodes are shown, for clarity. N.B. The cisterna chyli is more commonly formed by the intestinal lymph trunk and the left lumbar lymph trunk (rather than the right lumbar lymph trunk shown in this diagram).

clustered around its anterior unpaired visceral arteries and can be subdivided into coeliac, superior mesenteric and inferior mesenteric nodes. Efferent lymphatics from these nodes contribute to the formation of the intestinal lymph trunk.

Coeliac nodes

These drain lymph from nodes around the stomach, hilum of the spleen, porta hepatis, cystic duct, lesser omentum, portacaval nodes, peripancreatic nodes and pancreaticoduodenal nodes. They also receive lymph from superior and inferior mesenteric lymph nodes. Efferent lymphatics drain to the intestinal lymph trunk.

Superior mesenteric nodes

There is extensive overlap with the drainage territory of coeliac nodes, with subsidiary nodal groups that include peripancreatic, pancreaticoduodenal, portacaval, small bowel mesenteric, ileocolic, mesocolic and inferior mesenteric nodes. Efferent lymphatics drain directly to the intestinal lymph trunk or via coeliac nodes.

Inferior mesenteric nodes

These nodes drain lymph from the hindgut, which includes the distal transverse colon, descending and sigmoid colon, and rectum (including superior rectal, mesorectal and presacral nodes).

Lateral aortic groups

The lateral aortic (or para-aortic) nodes lie on either side of the abdominal aorta and inferior vena cava anterior to the medial margins of psoas major, diaphragmatic crura and sympathetic trunks. Constituent nodal groups that are recognized clinically include: retrocrural (posterior to the diaphragmatic crura at the aortic hiatus); left and right renal hilar; and aortocaval, paracaval, retrocaval and precaval nodes. Retro-aortic lymph nodes are also para-aortic and are, therefore, sometimes included within the lateral aortic group. The lateral aortic nodes drain into the paired lumbar lymph trunks, one on each side, which terminate directly or indirectly in the cisterna chyli and thoracic duct. Lymphatic connections exist between lateral aortic, pre-aortic, retro-aortic and contralateral lateral aortic nodes.

The lateral aortic nodes drain the deep layers of the body wall, retroperitoneal paired viscera (including the gonads) and the iliac nodal chains. Lymphatic drainage from the right testis is via lymphatics travelling with the gonadal vessels to the right para-aortic and aortocaval nodes at the level of the second lumbar vertebra, whereas the left testis drains to the left para-aortic nodes just inferior to the left renal vein (Paño et al 2011).

Iliac nodes

The paired iliac nodes are distributed around the common, external and internal iliac arteries and veins. Constituent groups include: common iliac, external iliac, internal iliac, circumflex iliac and obturator nodes. Obturator nodes are located near the obturator foramen and, along with the iliac nodes, are a common site of lymph node metastasis in prostate cancer. 'Pelvic side-wall' nodes, lying adjacent to the ilium but not associated with a large artery or vein, are also described in the clinical literature. The iliac nodes drain the pelvic viscera and walls, except for the ovaries and those parts of the rectum drained by superior rectal drainage pathways (see above). They also drain lymph from the inguinal nodes and lower limbs.

INNERVATION

The posterior abdominal wall contains the lumbar plexus and numerous autonomic plexuses and ganglia, which lie close to the abdominal aorta and its branches (**Fig. 62.14**).

LUMBAR PLEXUS

The lumbar ventral rami pass laterally into the posterior part of psoas major, anterior to the transverse processes of the lumbar vertebrae; the plexus lies in a coronal plane that is in line with the posterior part of the vertebral body at L1 but becomes more anterior in the lower lumbar spine (Moro et al 2003, Benglis et al 2009) (see Fig. 62.14; **Fig. 62.15**). The first four lumbar ventral rami, together with a contribution from the twelfth thoracic ventral ramus (the dorsolumbar nerve), form the lumbar plexus. Although there are many variations, the most common arrangement of the plexus is described here.

The first lumbar ventral ramus is joined by a branch from the twelfth thoracic ventral ramus, and these roots contribute to the formation of the iliohypogastric and ilioinguinal nerves, which run laterally on the posterior abdominal wall (see Fig. 62.14). A branch from the ventral ramus of L1 unites with a branch from the second lumbar ventral ramus to form the genitofemoral nerve. The second, third and most of the fourth lumbar ventral rami divide into ventral and dorsal divisions; the ventral divisions unite to form the obturator nerve, while most of the nerve fibres in the dorsal divisions form the femoral nerve. The remaining fibres from the fourth lumbar ventral ramus join the fifth lumbar ventral ramus to form the lumbosacral trunk, which descends to join the sacral plexus (p. 1229). Branches from the dorsal divisions of the second and third lumbar rami unite to form the lateral femoral cutaneous nerve (lateral cutaneous nerve of thigh). The accessory obturator nerve, when present, usually arises from the third and fourth ventral divisions. The lumbar plexus is supplied by branches from the lumbar vessels that supply psoas major.

The branches of the lumbar plexus are listed in **Table 62.1**.

Division of constituent ventral rami into ventral and dorsal branches is not as clear in the lumbar and sacral plexuses as it is in the brachial plexus. Lateral cutaneous branches of the twelfth thoracic and first lumbar ventral rami are drawn into the gluteal skin, but otherwise these nerves are similar to intercostal nerves. The second lumbar ventral ramus is more complex. It not only contributes substantially to the femoral and obturator nerves, but also has an anterior terminal branch (the genital branch of the genitofemoral) and a lateral cutaneous branch (which contributes to the lateral femoral cutaneous nerve and the femoral branch of the genitofemoral nerve). Anterior terminal branches of the third to fifth lumbar and first sacral rami are suppressed, but the corresponding branches of the second and third sacral rami supply perineal skin.

The furcal nerve is an independent nerve with its own ventral and dorsal rootlets most commonly arising alongside the L4 nerve root. Its branches contribute to the femoral and obturator nerves arising from the lumbar plexus and to the lumbosacral trunk, which joins the sacral plexus (Harshavardhana and Dabke 2014). The term furcal refers to its forked nature since it links the lumbar and sacral plexuses. Occasionally, the furcal nerve arises at the level of the third or the fifth lumbar nerve roots, in which case the sacral plexus is considered prefixed or postfixed, respectively.

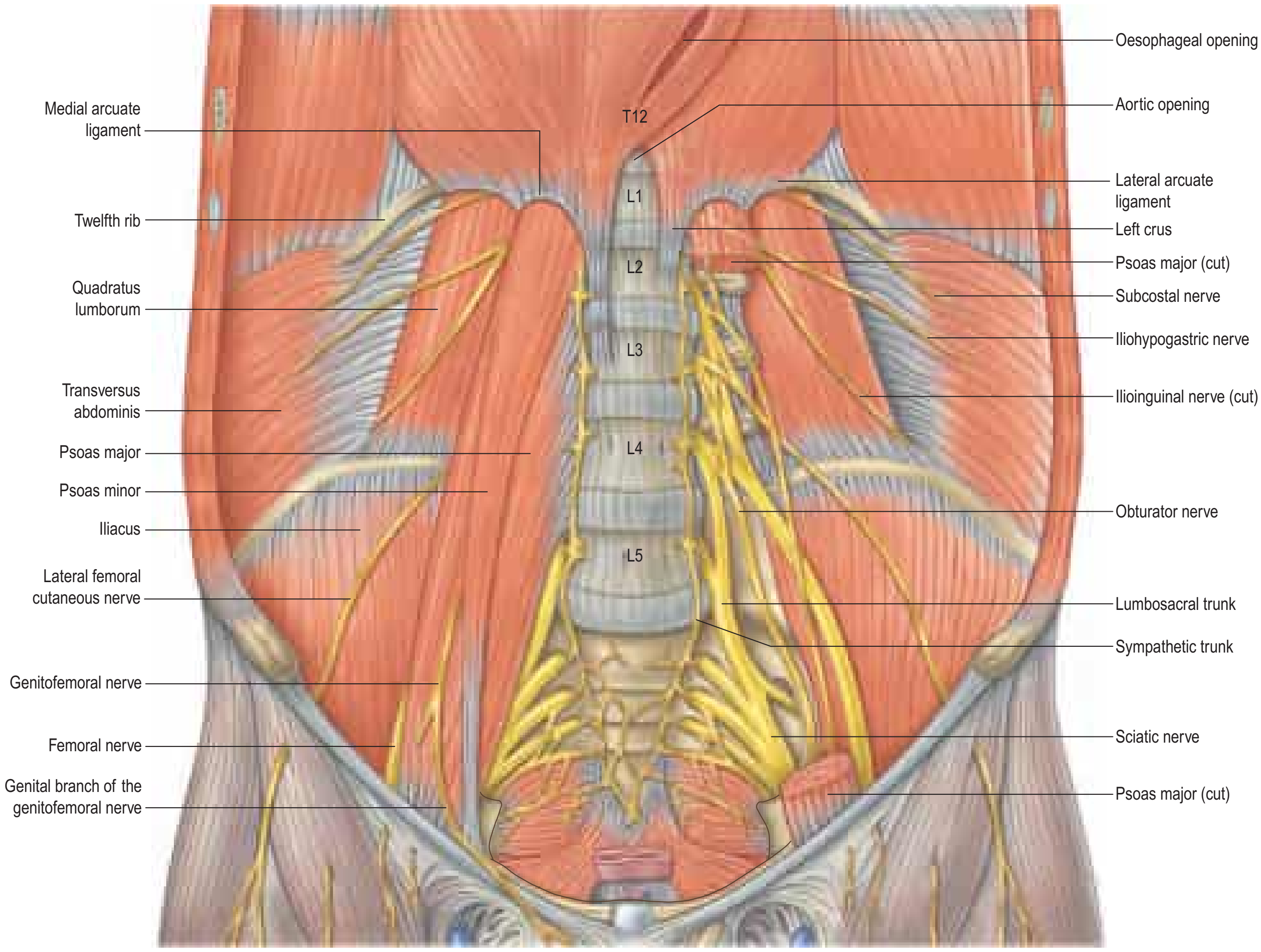

Fig. 62.14 Muscles and nerves of the posterior abdominal wall. The left psoas major has been removed to expose the origins of the lumbar plexus and quadratus lumborum.

Inflammatory processes, such as retrocaecal appendicitis on the right and diverticular abscess on the left, may occur in the posterior abdominal wall in the tissues immediately anterior to psoas major. These may irritate one or more branches of the lumbar plexus, causing pain or sensory disturbance in the distribution of the affected nerves, e.g. the skin of the thigh, hip or buttock.

Muscular branches

Small branches from the lumbar roots supply adjacent muscles such as psoas major and quadratus lumborum.

Iliohypogastric nerve

Distribution

The iliohypogastric nerve usually originates from the L1 ventral ramus but may arise wholly or in part from the T12 ventral ramus (Klaassen et al 2011). It emerges from the upper lateral border of psoas major, and crosses obliquely behind the lower renal pole on the anterior surface of quadratus lumborum. Above the iliac crest, it enters the posterior part of transversus abdominis and then runs forwards between transversus abdominis and internal oblique, which it supplies. It gives off a lateral cutaneous branch that pierces internal and external oblique above the iliac crest and supplies the posterolateral gluteal skin. It continues forwards, pierces the internal oblique approximately 3 cm medial and 1 cm inferior to the anterior superior iliac spine (in adults), and then penetrates the external oblique aponeurosis approximately 3 cm above and medial to the superficial inguinal ring to supply the suprapubic skin. The iliohypogastric nerve frequently connects with the ilioinguinal nerve and, less commonly, with the subcostal nerve (see Fig. 61.5).

Motor

The iliohypogastric nerve contributes motor nerves to transversus abdominis and internal oblique, including the conjoint tendon.

Sensory

The iliohypogastric nerve supplies sensory fibres to transversus abdominis, internal oblique and external oblique, and innervates the posterolateral gluteal and suprapubic skin.

Injury

The nerve is occasionally injured by a surgical incision in the right iliac fossa (e.g. during an inguinal hernia repair, open appendicectomy or trocar placement) but there is rarely any noticeable sensory loss because the suprapubic skin is innervated from several sources. Division of the iliohypogastric nerve above the anterior superior iliac spine may weaken the posterior wall of the inguinal canal and predispose to formation of a direct inguinal hernia.

Ilioinguinal nerve

Distribution

The ilioinguinal nerve usually originates from the L1 ventral ramus but may receive a contribution from T12 or L2 (Klaassen et al 2011). It emerges from the lateral border of psoas major, with or just inferior to the iliohypogastric nerve. It passes obliquely across quadratus lumborum and the upper part of iliacus and enters transversus abdominis about 3 cm medial and 4 cm inferior to the anterior superior iliac spine (in the adult); at this site, it is readily blocked by local anaesthetic. In this region, a branch may connect with the iliohypogastric or lateral femoral cutaneous nerve (see Fig. 61.5). It pierces internal oblique a little lower down, supplies it, and then traverses the inguinal canal superficial to the spermatic cord or round ligament. It emerges with the cord from the superficial inguinal ring and divides into terminal sensory branches.

Numerous anatomical variations are described (Al-dabbagh 2002, Ndiaye et al 2007). The ilioinguinal and genitofemoral nerves may interconnect within the inguinal canal and, consequently, each innervates the skin of the genitalia to a variable extent (Rab et al 2001, Cesmebasi et al 2015). The ilioinguinal nerve may pierce the external

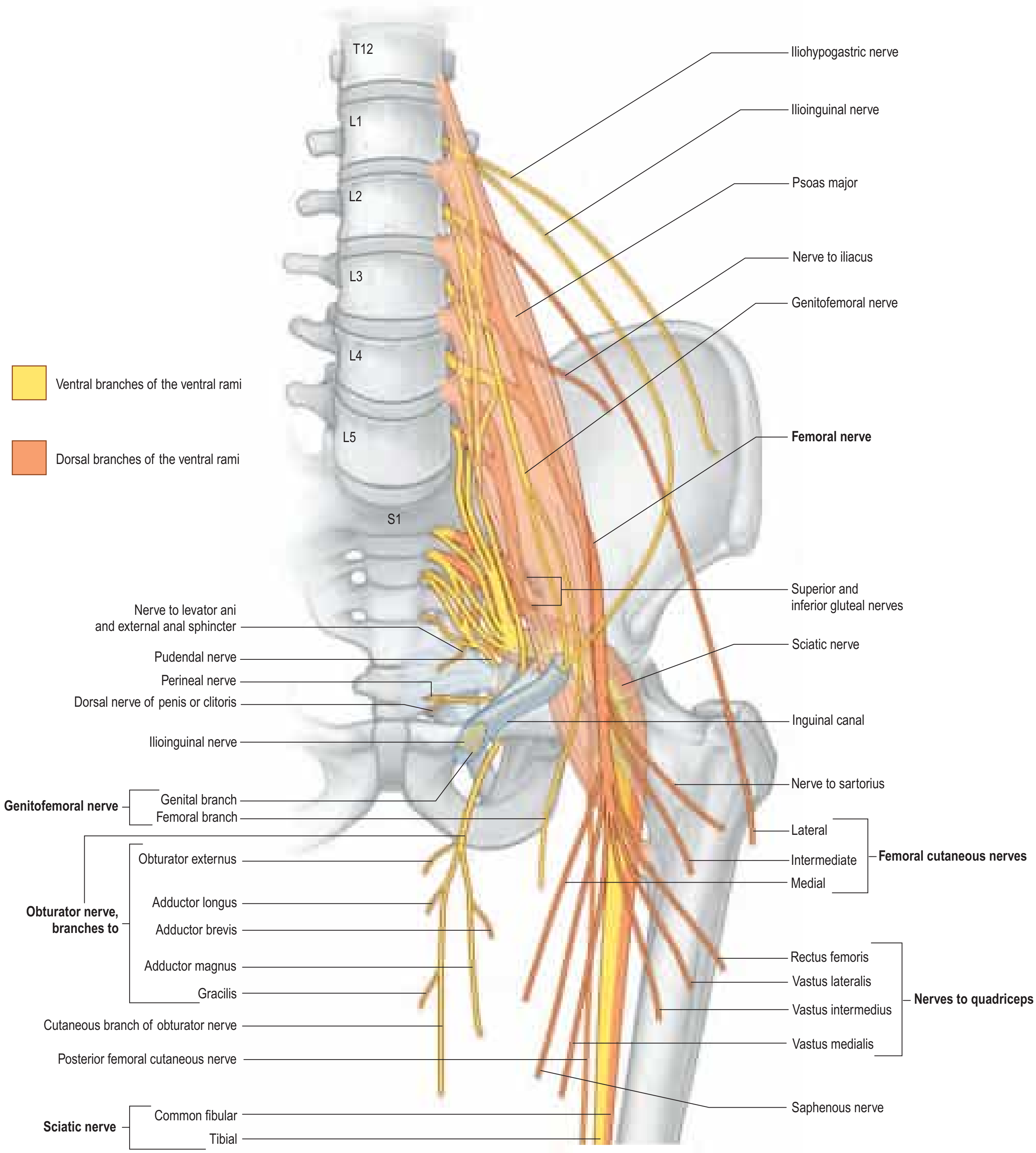

Fig. 62.15 The lumbar plexus and its branches.

Table 62.1 Branches of the lumbar plexus

Muscular	T12, L1–4
Iliohypogastric	L1
Ilioinguinal	L1
Genitofemoral	L1, L2
Lateral femoral cutaneous	L2, L3
Femoral	L2–4 dorsal divisions
Obturator	L2–4 ventral divisions
Accessory obturator	L3, L4

oblique aponeurosis proximal to the superficial inguinal ring or divide into terminal branches within the inguinal canal. It may form a common trunk with the iliohypogastric nerve as far as the midpoint of the inguinal canal. It may be very small or completely absent, in which case the iliohypogastric and genital branch of the genitofemoral nerves supplies its territory. Rarely, it may arise as a branch of the genitofemoral nerve.

Motor

The ilioinguinal nerve supplies motor nerves to transversus abdominis and internal oblique; these fibres are given off before the nerve enters the lateral end of the inguinal canal.

Sensory

The ilioinguinal nerve supplies sensory fibres to transversus abdominis and internal oblique. It innervates the skin of the proximal medial thigh and the skin over the root of the penis and upper part of the scrotum in males, or the skin covering the mons pubis and the adjoining labium majus in females.

Injury

The nerve may be injured or entrapped during inguinal surgery, particularly for inguinal hernia, leading to sensory disturbances and pain over the skin of the genitalia and upper medial thigh (Ndiaye et al 2007).

Genitofemoral nerve

Distribution

The genitofemoral nerve originates from the L1 and L2 ventral rami and is formed within the substance of psoas major. It descends obliquely forwards through the muscle to emerge on its anterior surface nearer the medial border, opposite the third or fourth lumbar vertebra (Moro et al 2003). It then descends beneath the peritoneum on psoas major, crosses obliquely behind the ureter, and divides into genital and femoral branches; it may divide close to its origin such that its branches emerge separately from psoas major. The genital branch crosses the lower part of the external iliac artery, enters the inguinal canal through the deep ring and accompanies the spermatic cord or round ligament. It exits the superficial inguinal ring, usually dorsal to the spermatic cord or round ligament, and supplies the cremaster muscle and skin of the external genitalia. The femoral branch descends lateral to the external iliac artery before crossing the deep circumflex iliac artery, to pass behind the inguinal ligament (occasionally, through it) (Rab et al 2001) and enter the femoral sheath lateral to the femoral artery. It pierces the anterior layer of the femoral sheath and fascia lata, and supplies the skin of the upper part of the femoral triangle. It may connect with the lateral femoral cutaneous and intermediate femoral cutaneous nerves.

Motor

The genitofemoral nerve innervates cremaster via the genital branch.

Sensory

The genitofemoral nerve innervates the skin of the scrotum in males, or that of the mons pubis and labium majus in females, via its genital branch; there is considerable overlap and variability with the cutaneous distribution of the ilioinguinal nerve (Cesmebasi et al 2015). It also innervates the anteromedial skin of the thigh via its femoral branch.

The genitofemoral nerve is also understood to play a critical role in inguinoscrotal descent of the developing testis (Hutson et al 2015).

Injury

Like the ilioinguinal nerve, the genital branch may be injured during inguinal surgery (open and laparoscopic), leading to neuralgic pain (Cesmebasi et al 2015).

Femoral nerve

The femoral nerve descends through psoas major and emerges on or under its lateral border, about 4 cm above the inguinal ligament (Moore and Stringer 2011). It passes between psoas major and iliacus deep to the iliac fascia and runs posterior to the inguinal ligament into the thigh. It gives off branches that supply iliacus and pectineus, and sends sensory fibres to the femoral artery. Posterior to the inguinal ligament, it lies lateral to the femoral artery and sheath. The further course and distribution are described on page 1372.

Lateral femoral cutaneous nerve (lateral cutaneous nerve of the thigh)

The lateral femoral cutaneous nerve (lateral cutaneous nerve of the thigh) is usually derived from the ventral rami of L2 and 3, but variable contributions from L1 to L3 are described (de Ridder et al 1999). It emerges from the posterolateral border of psoas major and crosses iliacus obliquely towards the anterior superior iliac spine. It supplies sensory fibres to the parietal peritoneum in the iliac fossa. On the right, the nerve passes posterolateral to the caecum, separated from it by the iliac fascia and peritoneum. The left nerve passes behind the lower part of the descending colon. Both nerves usually pass behind the inguinal ligament about 1–2 cm medial to the anterior superior iliac spine; occasionally, they pass through or, rarely, anterior to the ligament (Ray et al 2010). Occasionally, the nerve lies anterior or superior to the anterior superior iliac spine as it enters the thigh. In the thigh, the lateral femoral cutaneous nerve usually passes anterior or lateral to sartorius but may pierce the muscle. The further course and distribution are described on page 1371.

Obturator nerve

The obturator nerve descends within the substance of psoas major to emerge from its posteromedial border near the L5 vertebra (Kirchmair et al 2008). It passes posterior to the common iliac vessels and lateral to the internal iliac vessels. It then descends on the lateral wall of the pelvis attached to the fascia over obturator internus and lies anterosuperior to the obturator vessels before running into the obturator foramen to enter the thigh. It has no branches in the abdomen or pelvis. The further course and distribution are described on page 1372.

Accessory obturator nerve

The accessory obturator nerve is sometimes present, more often on the left (Katritsis et al 1980). It is usually formed by the ventral rami of L3 and L4. It emerges from the medial border of psoas major and runs along the posterior surface of the superior pubic ramus posterior to pectineus, where it gives off branches to supply pectineus and the hip joint, and may join with the anterior branch of the obturator nerve (p. 1372).

LUMBAR SYMPATHETIC SYSTEM

The lumbar part of each sympathetic trunk usually contains four interconnected ganglia lying in extraperitoneal connective tissue on the anterolateral aspects of the lumbar vertebrae along the medial margin of psoas major (see Fig. 62.14). Superiorly, it is continuous with the thoracic sympathetic trunk posterior to the medial arcuate ligament at the L1/2 vertebral level (Feigl et al 2011). Inferiorly, it passes posterior to the common iliac vessels and is continuous with the sacral sympathetic trunk. On the right side, it lies posterior to the inferior vena cava; on the left, it is posterior to the lateral aortic lymph nodes. It is anterior to most of the lumbar vessels but may pass behind some lumbar veins.

The first, second and, sometimes, the third lumbar ventral rami are each connected to the lumbar sympathetic trunk by a white ramus communicans. All lumbar ventral rami are joined near their origins by long, slender grey rami communicantes from the four lumbar sympathetic ganglia. Their arrangement is irregular: one ganglion may give rami to two or three lumbar ventral rami, one lumbar ventral ramus may receive rami from two ganglia, or grey rami may leave the sympathetic trunk between ganglia (Murata et al 2003).

The lumbar sympathetic trunks are vulnerable during retroperitoneal nodal dissection and their injury can impair seminal emission and lead to retrograde ejaculation.

LUMBAR PARASYMPATHETIC SYSTEM

The parasympathetic supply to the abdominal viscera is provided by the vagus nerve to the coeliac and superior mesenteric plexuses, and by pelvic splanchnic nerves that are distributed through the hypogastric and inferior mesenteric plexuses (Ch. 59).

PARA-AORTIC BODIES

The para-aortic bodies (also known as paraganglia or, collectively, as the organ of Zuckerkandl) are collections of neural crest-derived chromaffin tissue found in close relation to the aortic autonomic plexuses. They are relatively large in the fetus, where they may have a role in maintaining blood pressure by catecholamine secretion. They reach a maximum size at around 3 years of age, and have usually regressed by adulthood. They are usually found as a pair of bodies lying anterolateral to the aorta in the region of the inferior mesenteric and superior hypogastric plexuses, but multiple smaller collections may be present. Occasionally, they are found as high as the coeliac plexus or as low as the inferior hypogastric plexus in the pelvis, or are closely applied to the sympathetic ganglia of the lumbar chain. Scattered cells that persist into adulthood may, rarely, be the sites of paraganglioma (extra-adrenal phaeochromocytoma) (Subramanian and Maker 2006).

KEY REFERENCES

Congdon ED, Blumberg R, Henry W 1942 Fasciae of fusion and elements of the fused enteric mesenteries in the human adult. Am J Anat 70: 251–9.

Culligan K, Remzi FH, Soop M et al 2013 Review of nomenclature in colonic surgery – proposal of a standardised nomenclature based on mesocolic anatomy. Surgeon 11:1–5.
Making sense of mesocolic anatomy based on findings from modern investigative techniques.

Dodds WJ, Darweesh RMA, Lawson TL et al 1986 The retroperitoneal spaces revisited. AJR Am J Roentgenol 147:1155–61.
An important radiologic perspective on the retroperitoneal 'spaces'.

Klaassen Z, Marshall E, Tubbs RS et al 2011 Anatomy of the ilioinguinal and iliohypogastric nerves with observations of their spinal nerve contributions. Clin Anat 24:454–61.

Ndiaye A, Diop M, Ndoye JM et al 2007 Anatomical basis of neuropathies and damage to the ilioinguinal nerve during repairs of groin hernias (about 100 dissections). Surg Radiol Anat 29:675–81.
Together with Klaassen et al 2011, these two papers provide a useful description of the anatomy of the ilioinguinal and iliohypogastric nerves and underline the importance of the ilioinguinal nerve in clinical practice.

Loukas M, Wartmann CT, Louis RG Jr et al 2007b Cisterna chyli: a detailed anatomic investigation. Clin Anat 20:683–8.
This article revises traditional concepts about the cisterna chyli and its formation.

Mirjalili SA, McFadden SL, Buckenham T et al 2012b A reappraisal of adult abdominal surface anatomy. Clin Anat 25:844–50.
A paper that challenges some traditional surface anatomical landmarks by investigating results from analysis of CT scans in living supine adults.

Willard FH, Vleeming A, Schuenke MD et al 2012 The thoracolumbar fascia: anatomy, function and clinical considerations. J Anat 221:507–36.
An evidence-based review of the complex and clinically important thoracolumbar fascia.

REFERENCES

Al-dabbagh AK 2002 Anatomical variations of the inguinal nerves and risks of injury in 110 hernia repairs. Surg Radiol Anat 24:102–7.

Allan PL 2006 The aorta and the inferior vena cava. In: Allan PL, Dubbins PA, Pozniak MA et al (eds) Clinical Doppler Ultrasound, 2nd ed. Edinburgh: Elsevier, Churchill Livingstone.

Amin M, Blandford AT, Polk HC Jr 1976 Renal fascia of Gerota. Urology 7:1–3.

Ang WC, Doyle T, Stringer MD 2013 Left-sided and duplicate inferior vena cava: a case series and review. Clin Anat 26:990–1001.

Arslan M, Comert A, Acar HI et al 2011 Surgical view of the lumbar arteries and their branches: an anatomical study. Neurosurgery 68:16–22.

Barber B, Horton A, Patel U 2012 Anatomy of the origin of the gonadal veins on CT. J Vasc Interv Radiol 23:211–15.

Benglis DM, Vanni S, Levi AD 2009 An anatomical study of the lumbosacral plexus as related to the minimally invasive transpsoas approach to the lumbar spine. J Neurosurg Spine 10:139–44.

Bergman RA, Afifi AK, Miyauchi R 2014 Illustrated Encyclopedia of Human Anatomic Variation: Opus II: Cardiovascular System: Arteries: Abdomen: Superior Mesenteric Artery. Available at: http://www.anatomyatlases.org/AnatomicVariants/Cardiovascular/Text/Arteries/MesentericSuperior.shtml. Accessed 11 May 2015.

Biglioli P, Roberto M, Cannata A et al 2004 Upper and lower spinal cord blood supply: the continuity of the anterior spinal artery and the relevance of the lumbar arteries. J Thorac Cardiovasc Surg 127:1188–92.

Cesmebasi A, Du Plessis M, Iannatuono M et al 2014 A review of the anatomy and clinical significance of adrenal veins. Clin Anat 27:1253–63.

Cesmebasi A, Yadav A, Gielecki J et al 2015 Genitofemoral neuralgia: a review. Clin Anat 28:128–35.

Chesbrough RM, Burkhard TK, Martinez AJ et al 1989 Gerota versus Zuckerkandl: the renal fascia revisited. Radiology 173:845–6.

Coffin A, Boulay-Coletta I, Sebbag-Sfez D et al 2015 Radioanatomy of the retroperitoneal space. Diagn Interv Imaging 96:171–86.

Congdon ED, Edson JN 1941 The cone of renal fascia in the adult white male. Anat Rec 80:289–313.

Congdon ED, Blumberg R, Henry W 1942 Fasciae of fusion and elements of the fused enteric mesenteries in the human adult. Am J Anat 70:251–9.

Culligan K, Remzi FH, Soop M et al 2013 Review of nomenclature in colonic surgery – proposal of a standardised nomenclature based on mesocolic anatomy. Surgeon 11:1–5.
Making sense of mesocolic anatomy based on findings from modern investigative techniques.

Culligan K, Walsh S, Dunne C et al 2014 The mesocolon: a histological and electron microscopic characterization of the mesenteric attachment of the colon prior to and after surgical mobilization. Ann Surg 260:1048–56.

de Ridder VA, de Lange S, Popta JV 1999 Anatomical variations of the lateral femoral cutaneous nerve and the consequences for surgery. J Orthop Trauma 13:207–11.

Dodds WJ, Darweesh RMA, Lawson TL et al 1986 The retroperitoneal spaces revisited. AJR Am J Roentgenol 147:1155–61.
An important radiologic perspective on the retroperitoneal 'spaces'.

Feigl GC, Kastner M, Ulz H et al 2011 Topography of the lumbar sympathetic trunk in normal lumbar spines and spines with spondylophytes. Br J Anaesth 106:260–5.

Fleischmann D, Hastie TJ, Dannegger FC et al 2001 Quantitative determination of age-related geometric changes in the normal abdominal aorta. J Vasc Surg 33:97–105.

Golub RM, Parsons RE, Sigel B et al 1992 A review of venous collaterals in inferior vena cava obstruction. Clin Anat 5:441–51.

Gwon DI, Ko GY, Yoon HK et al 2007 Inferior phrenic artery: anatomy, variations, pathologic conditions, and interventional management. Radiographics 27:687–705.

Harshavardhana NS, Dabke HV 2014 The furcal nerve revisited. Orthop Rev (Pavia) 6:5428.

Hebbard P, Ivanusic J, Sha S 2011 Ultrasound-guided supra-inguinal fascia iliaca block: a cadaveric evaluation of a novel approach. Anaesthesia 66:300–5.

Hutson JM, Li R, Southwell BR et al 2015 Regulation of testicular descent. Pediatr Surg Int 31:317–25.

Jasper A, Harshe G, Keshava SN et al 2014 Evaluation of normal abdominal aortic diameters in the Indian population using computed tomography. J Postgrad Med 60:57–60.

Katritsis E, Anagnostopoulou S, Papadopoulos N 1980 Anatomical observations on the accessory obturator nerve (based on 1000 specimens). Anat Anz 148:440–5.

Kiil BJ, Rozen WM, Pan WR et al 2009 The lumbar artery perforators: a cadaveric and clinical anatomical study. Plast Reconstr Surg 123:1229–38.

Kimura W, Yano M, Sugawara S et al 2010 Spleen-preserving distal pancreatectomy with conservation of the splenic artery and vein: techniques and its significance. J Hepatobiliary Pancreat Sci 17:813–23.

Kirchmair L, Lirk P, Colvin J et al 2008 Lumbar plexus and psoas major muscle: not always as expected. Reg Anesth Pain Med 33:109–14.

Klaassen Z, Marshall E, Tubbs RS et al 2011 Anatomy of the ilioinguinal and iliohypogastric nerves with observations of their spinal nerve contributions. Clin Anat 24:454–61.
Together with Ndiaye et al 2007, these two papers provide a useful description of the anatomy of the ilioinguinal and iliohypogastric nerves and underline the importance of the ilioinguinal nerve in clinical practice.

Kneeland JB, Auh YH, Rubenstein WA et al 1987 Perirenal spaces: CT evidence for communication across the midline. Radiology 164:657–64.

Lee MW, McPhee RW, Stringer MD 2008 An evidence-based approach to human dermatomes. Clin Anat 21:363–73.

Lolis E, Panagouli E, Venieratos D 2011 Study of the ascending lumbar and iliolumbar veins: surgical anatomy, clinical implications and review of the literature. Ann Anat 193:516–29.

Loukas M, Hullett J, Wagner T 2005a Clinical anatomy of the inferior phrenic artery. Clin Anat 18:357–65.

Loukas M, Louis RG Jr, Hullett J et al 2005b An anatomical classification of the variations of the inferior phrenic vein. Surg Radiol Anat 27:566–74.

Loukas M, Pinyard J, Vaid S et al 2007a Clinical anatomy of celiac artery compression syndrome: a review. Clin Anat 20:612–17.

Loukas M, Wartmann CT, Louis RG Jr et al 2007b Cisterna chyli: a detailed anatomic investigation. Clin Anat 20:683–8.
This article revises traditional concepts about the cisterna chyli and its formation.

Mirjalili SA, Hale SJ, Buckenham T et al 2012a A reappraisal of adult thoracic surface anatomy. Clin Anat 25:827–34.

Mirjalili SA, McFadden SL, Buckenham T et al 2012b A reappraisal of adult abdominal surface anatomy. Clin Anat 25:844–50.
A paper that challenges some traditional surface anatomical landmarks by investigating results from analysis of CT scans in living supine adults.

Moeller RB, Reif E 2000 Normal Findings in CT and MRI. Rome: CIC Edizioni Internazionali.

Moore AE, Stringer MD 2011 Iatrogenic femoral nerve injury: a systematic review. Surg Radiol Anat 33:649–58.

Morita S, Kimura T, Masukawa A et al 2007 Flow direction of ascending lumbar veins on magnetic resonance angiography and venography: would 'descending lumbar veins' be a more precise name physiologically? Abdom Imaging 32:749–53.

Moro T, Kikuchi S, Konno S et al 2003 An anatomic study of the lumbar plexus with respect to retroperitoneal endoscopic surgery. Spine (Phila Pa 1976) 28:423–8.

Moussallem CD, Abou Hamad I, El-Yahchouchi CA et al 2012 Relationship of the lumbar lordosis angle to the abdominal aortic bifurcation and inferior vena cava confluence levels. Clin Anat 25:866–71.

Murata Y, Takahashi K, Yamagata M et al 2003 Variations in the number and position of human lumbar sympathetic ganglia and rami communicantes. Clin Anat 16:108–13.

Ndiaye A, Diop M, Ndoye JM et al 2007 Anatomical basis of neuropathies and damage to the ilioinguinal nerve during repairs of groin hernias (about 100 dissections). Surg Radiol Anat 29:675–81.
Together with Klaassen et al 2011, these two papers provide a useful description of the anatomy of the ilioinguinal and iliohypogastric nerves and underline the importance of the ilioinguinal nerve in clinical practice.

Needleman L 2006 Measurements of the abdominal aorta. In: Goldberg BB, McGahan JP (eds) Atlas of Ultrasound Measurements, 2nd ed. Boston: Elsevier, Mosby.

Niggemann P, Förg A, Grosskurth D et al 2010 Postural effect on the size of the cisterna chyli. Lymphat Res Biol 8:193–7.

Panagouli E, Venieratos D, Lolis E et al 2013 Variations in the anatomy of the celiac trunk: a systematic review and clinical implications. Ann Anat 195:501–11.

Paño B, Sebastià C, Buñesch L et al 2011 Pathways of lymphatic spread in male urogenital pelvic malignancies. Radiographics 31:135–60.

Phang K, Bowman M, Phillips A et al 2014 Review of thoracic duct anatomical variations and clinical implications. Clin Anat 27:637–44.

Phillips S, Mercer S, Bogduk N 2008 Anatomy and biomechanics of quadratus lumborum. Proc Inst Mech Eng H 222:151–9.

Rab M, Ebmer J, Dellon AL 2001 Anatomic variability of the ilioinguinal and genitofemoral nerve: implications for the treatment of groin pain. Plast Reconstr Surg 108:1618–23.

Ranson C, Burnett A, O'Sullivan P et al 2008 The lumbar paraspinal muscle morphometry of fast bowlers in cricket. Clin J Sport Med 18:31–7.

Ray B, D'Souza AS, Kumar B et al 2010 Variations in the course and microanatomical study of the lateral femoral cutaneous nerve and its clinical importance. Clin Anat 23:978–84.

Rogers IS, Massaro JM, Truong QA et al 2013 Distribution, determinants, and normal reference values of thoracic and abdominal aortic diameters by computed tomography (from the Framingham Heart Study). Am J Cardiol 111:1510–16.

Spentzouris G, Zandian A, Cesmebasi A et al 2014 The clinical anatomy of the inferior vena cava: a review of common congenital anomalies and considerations for clinicians. Clin Anat 27:1234–43.

Stamatiou D, Skandalakis JE, Skandalakis LJ et al 2009 Lumbar hernia: surgical anatomy, embryology, and technique of repair. Am Surg 75: 202–7.

Subramanian A, Maker VK 2006 Organs of Zuckerkandl: their surgical significance and a review of a century of literature. Am J Surg 192: 224–34.

Takayama T, Yamanouchi D 2013 Aneurysmal disease: the abdominal aorta. Surg Clin North Am 93:877–91.

Toni R, Mosca S, Favero L et al 1988 Clinical anatomy of the suprarenal arteries: a quantitative approach by aortography. Surg Radiol Anat 10: 297–302.

Tourani SS, Taylor GI, Ashton MW 2013 Anatomy of the superficial lymphatics of the abdominal wall and the upper thigh and its implications in lymphatic microsurgery. J Plast Reconstr Aesthet Surg 66:1390–5.

Wegener OH 1992 Whole body computed tomography, 2nd ed. Boston: Blackwell Scientific.

Willard FH, Vleeming A, Schuenke MD et al 2012 The thoracolumbar fascia: anatomy, function and clinical considerations. J Anat 221:507–36.
An evidence-based review of the complex and clinically important thoracolumbar fascia.

Winston CB, Lee NA, Jarnagin WR et al 2007 CT angiography for delineation of celiac and superior mesenteric artery variants in patients undergoing hepatobiliary and pancreatic surgery. AJR Am J Roentgenol 189: W13–19.